



Science and
Technology
Facilities Council

Development and Quality control of Novel Scintillating Tiles for Super MuSR.

Ben Thompson

Outline:

1. Introduction

- What is the ISIS Neutron and Muon Source
- Production of Muons

2. MuSR & Super MuSR

- μ SR essentials
- MuSR vs Super MuSR
- Positron detection in super MuSR
- Tile production procedure

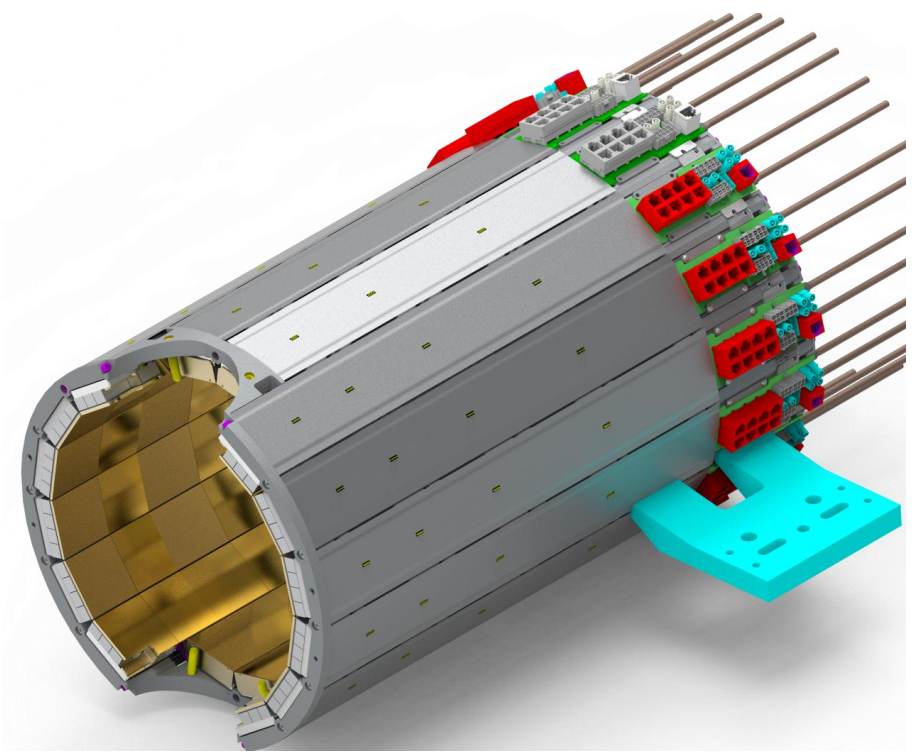
3. Experimental & Results

- Determining an acceptance threshold
- Tile Measurement Procedure
- Tile testing setup
- 30mm, 63mm, 105mm, & 210mm tile results

4. Conclusions

- Conclusion and Future Work
- Questions

This one is okay



Introduction

What is ISIS

- Pulsed Spallation Source
- Over 30 instruments
- 1200 Experiments
- 600 Publications

What does ISIS Study?

- Advanced materials
- Chemistry and catalysis
- Life sciences and healthcare
- Environment and geology

Production of Muons

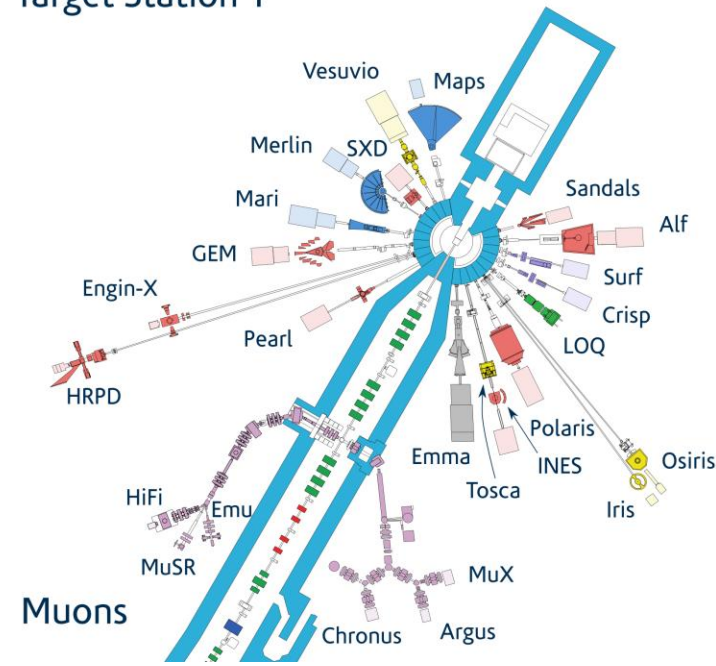
- RFQ and Linac
- 800MeV Synchrotron accelerates protons to 84% the speed of light
- Collide with carbon target
- Polarised Muons Selected
- Guided into the sample

- Quite a lot of information speak more of it this is across three slides, reduce it!

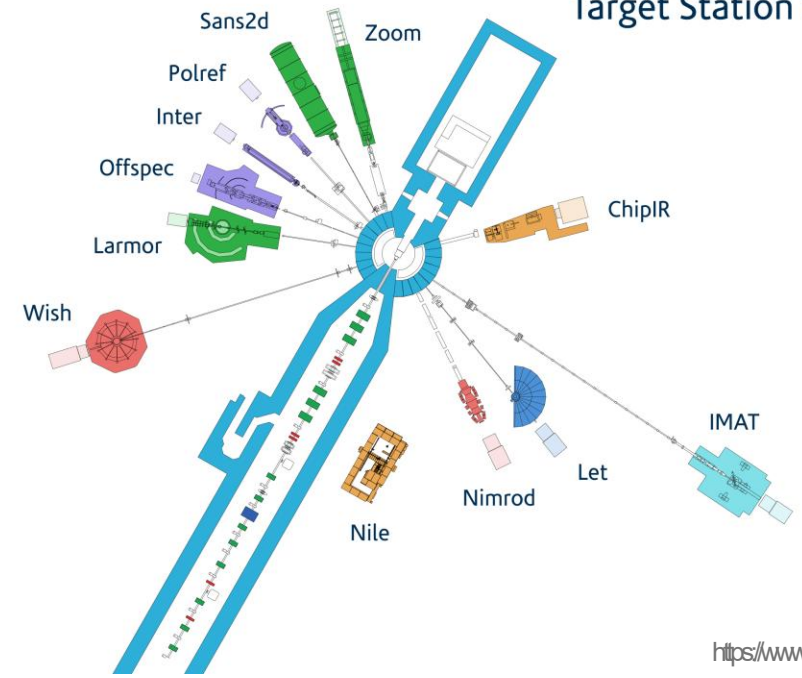
- Delete things that you can just say, no need to include figure



Target Station 1



Target Station 2



Introduction

What is ISIS

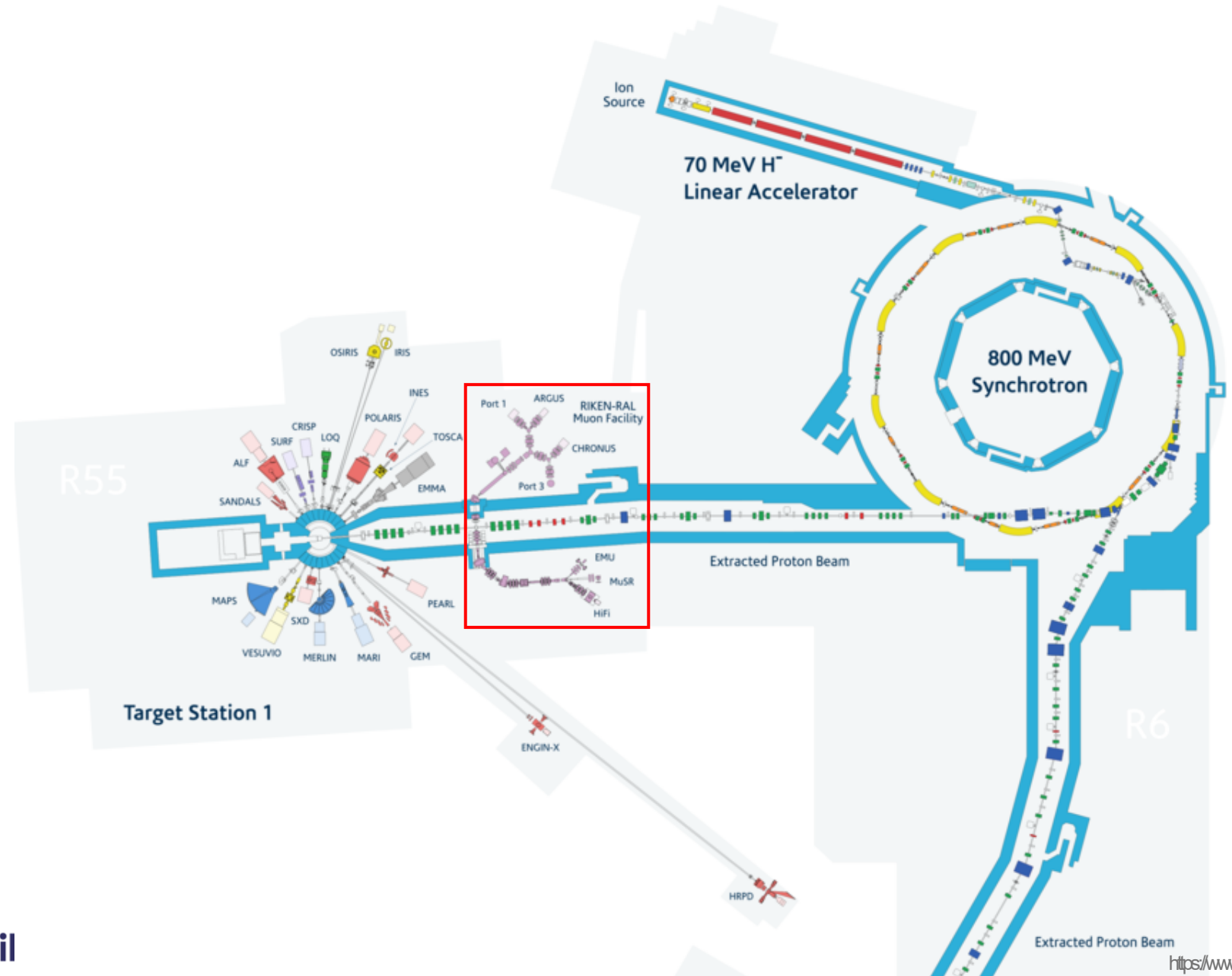
- Pulsed Spallation Source
- 37 Instruments
- world leading research
- 1200 Experiments
- 600 Publications

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Introduction

What is ISIS

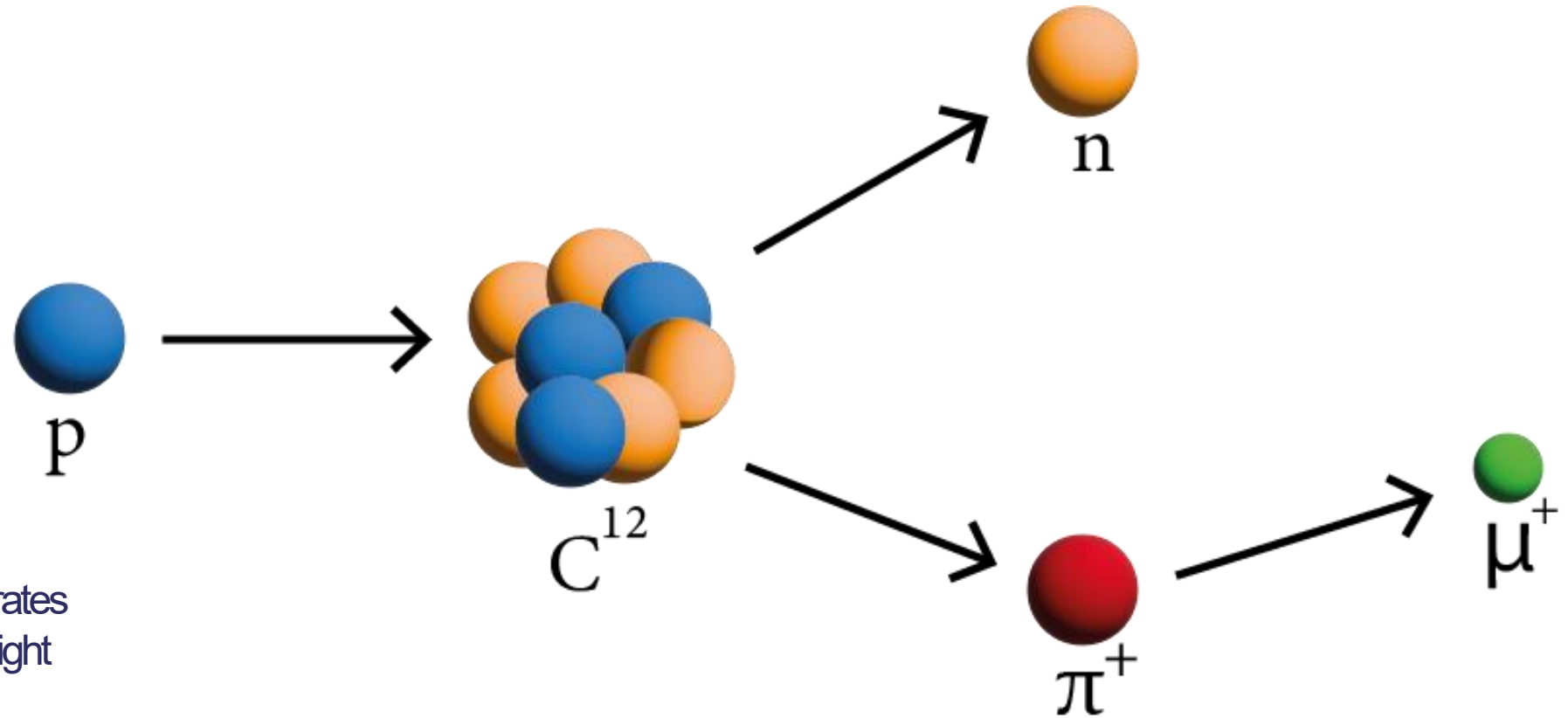
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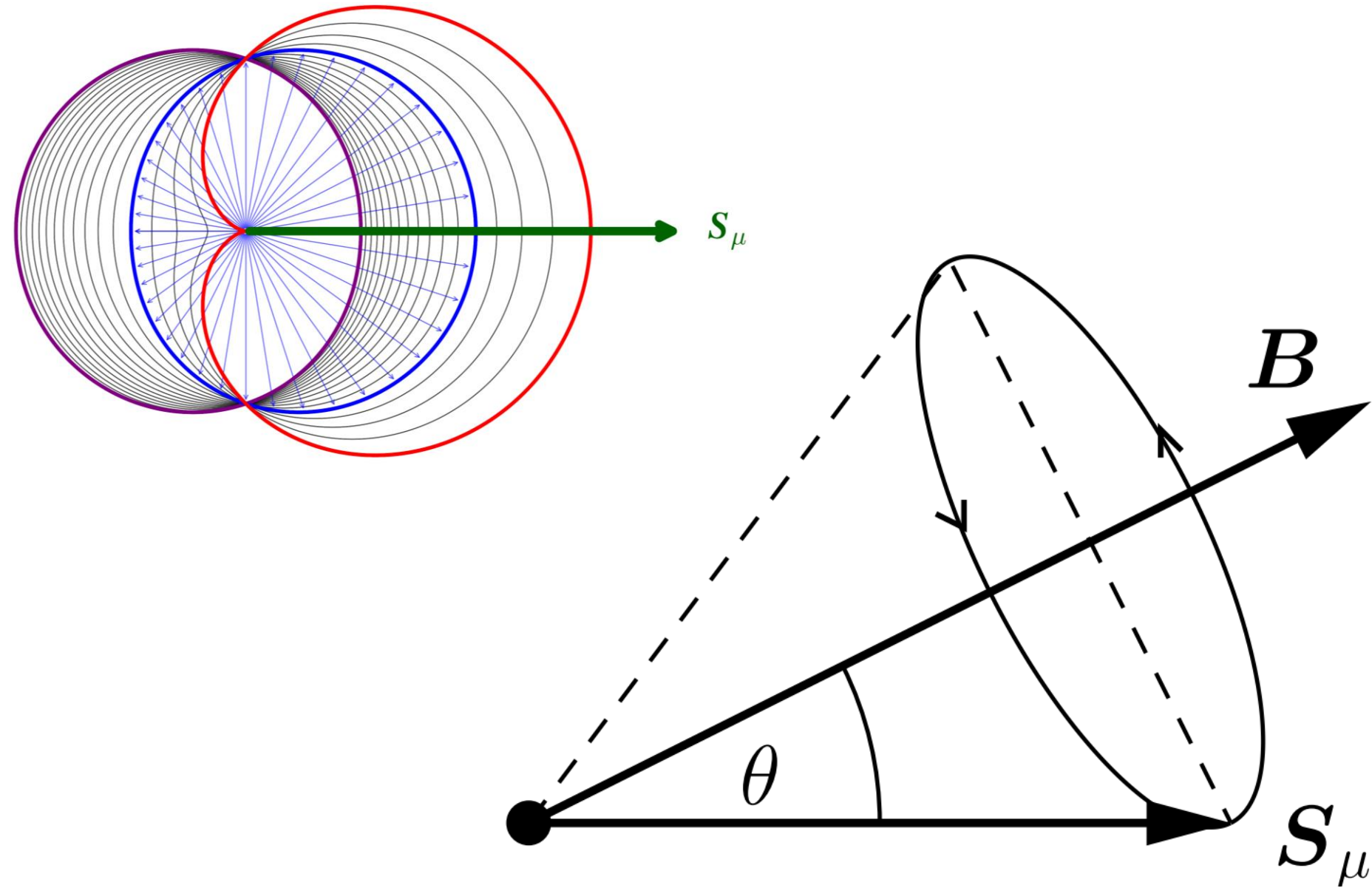
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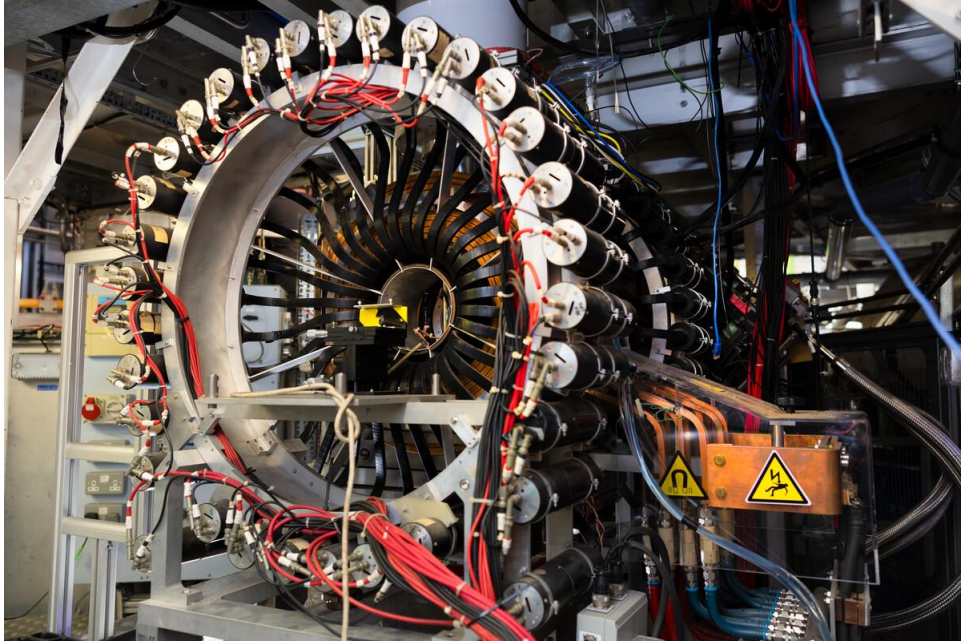


μ SR essentials

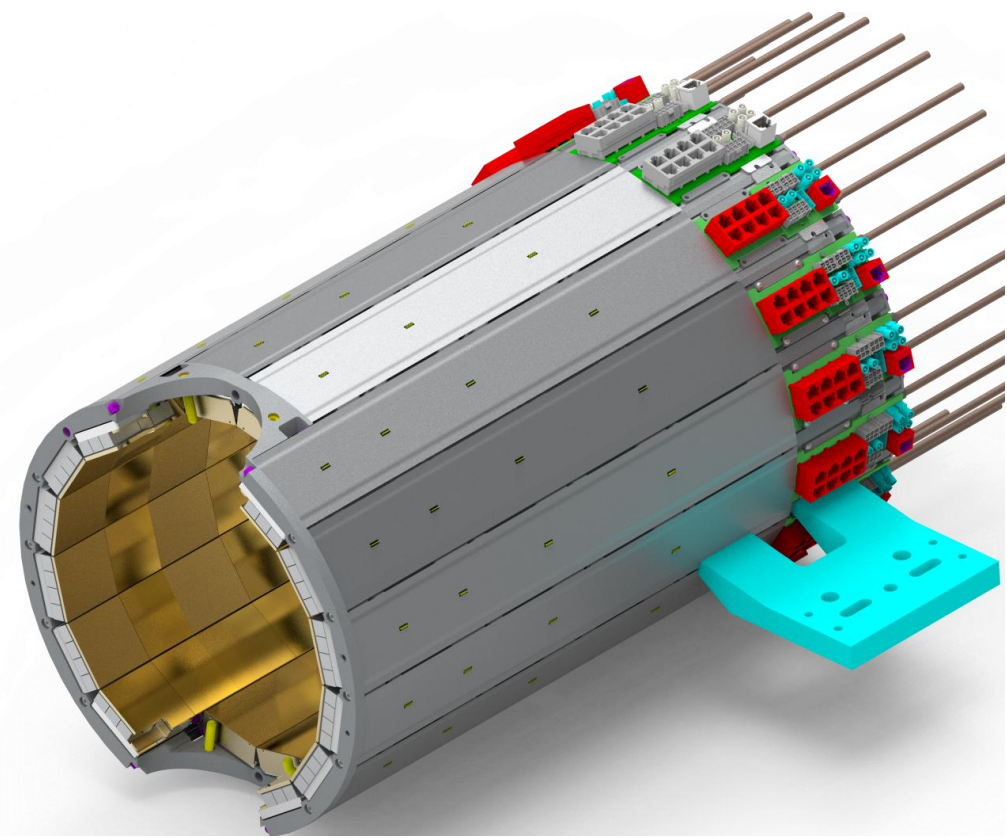
- Spin polarised muons arrive in the sample
- Muon spin precess around the local magnetic field
- $\omega = \gamma B$ allows determination of magnetic field strength
- Muon decay preferentially in direction of muon spin
- Surrounding the sample with detectors allows for the spin polarization and precession to be measured with respect to time
- This reveals the magnetic field distribution of the sample
- Tutor is not a muon scientist, what does the asymmetry stuff (a_0) is it important for the talk- no it is not.
- Remove some more inform



MuSR vs Super MuSR



- 64 detector elements
- PMTs
- Plastic Scintillator and Perspex to collect light
- Not much here, you were wondering about staves
- Do people waste time about what a stave is.
- Can andrea see the staves and stuff.

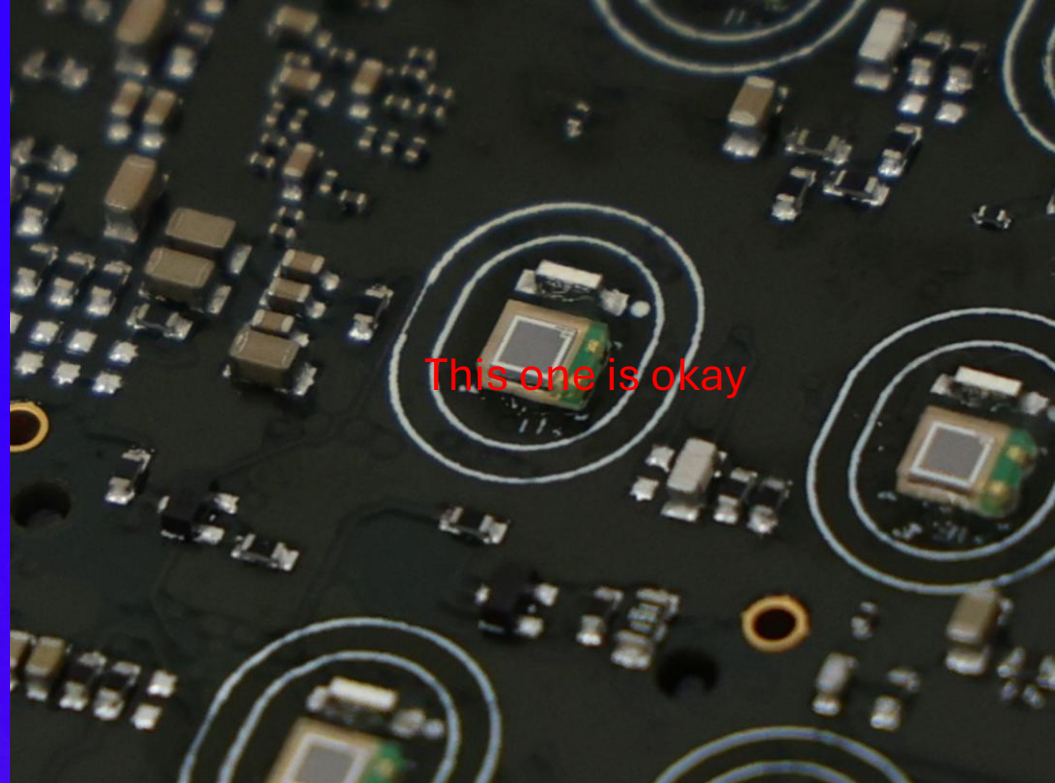


- 960 detector elements
- 6 -10x increase in temporal resolution
- 20x increase in data collection
- SiPMs
- 32 Staves in total, each made of 32 detector elements
- Improved time resolution allows for greater range of internal fields to be measured, facilitating more materials that can be investigated
- Higher data collection rates will allow for better battery material research
- Facilitates high statistic measurements in shorter timeframes

Positron Detection in Super MuSR

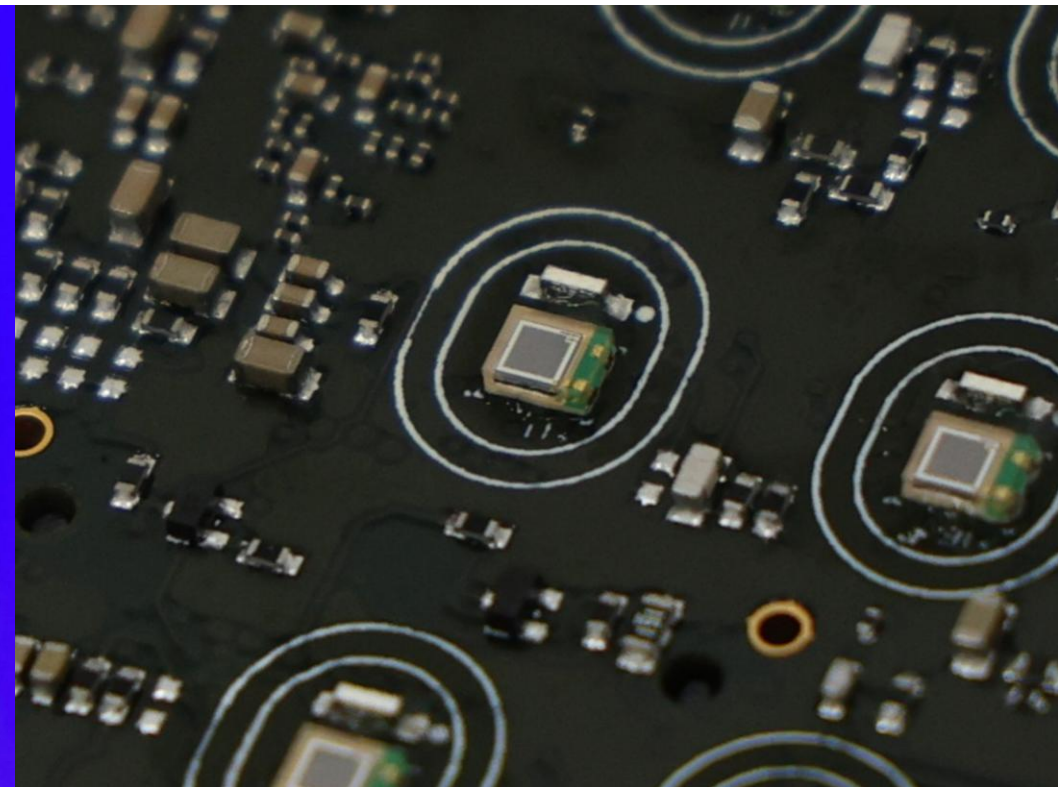
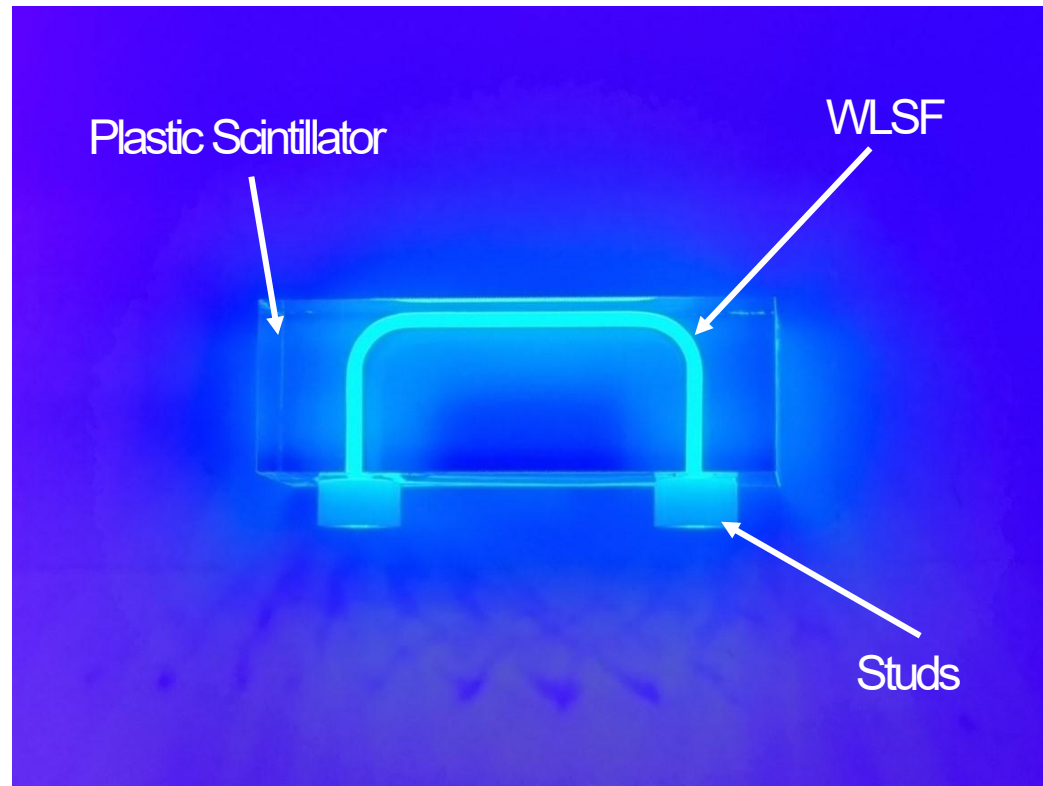


- Plastic Scintillator
- WLSF
- Bound by clear epoxy resin
- Studs for alignment



- Pixelated detectors
- Geiger mode in each pixel (either on or off)
- Similar Gain to the PMT
- Magnetically immune

Positron Detection in Super MuSR



- Plastic Scintillator
- Wavelength shifting fibre that absorbs light from scintillator and directs it to the sipms (WLSF)
- Bound by clear epoxy resin

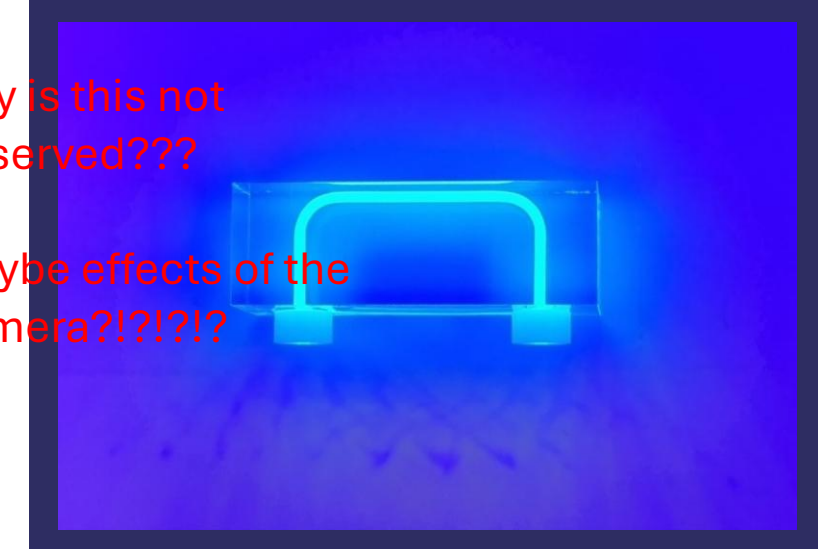
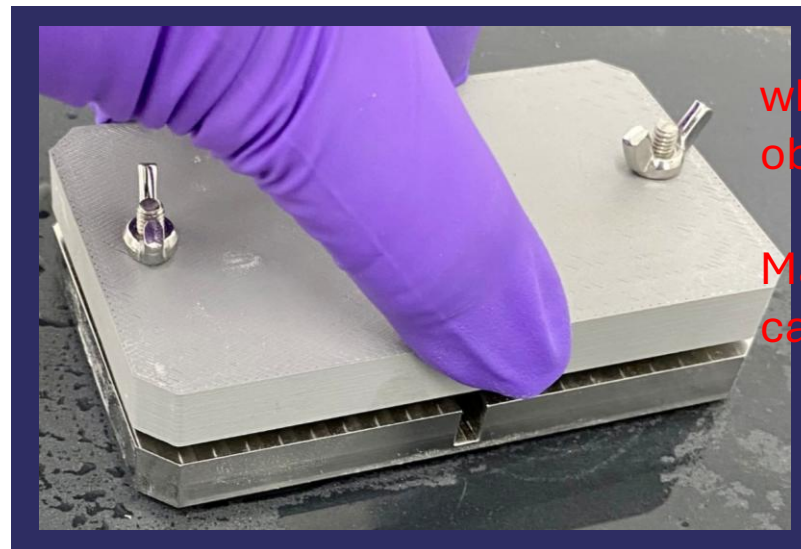
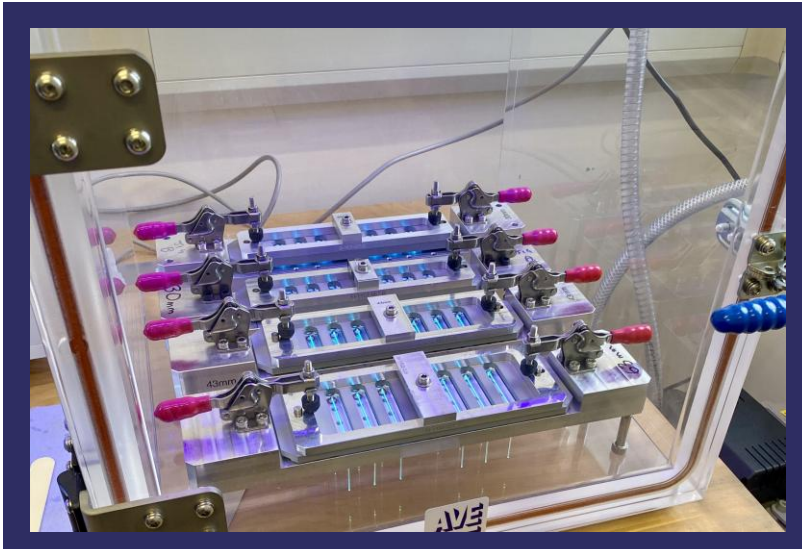
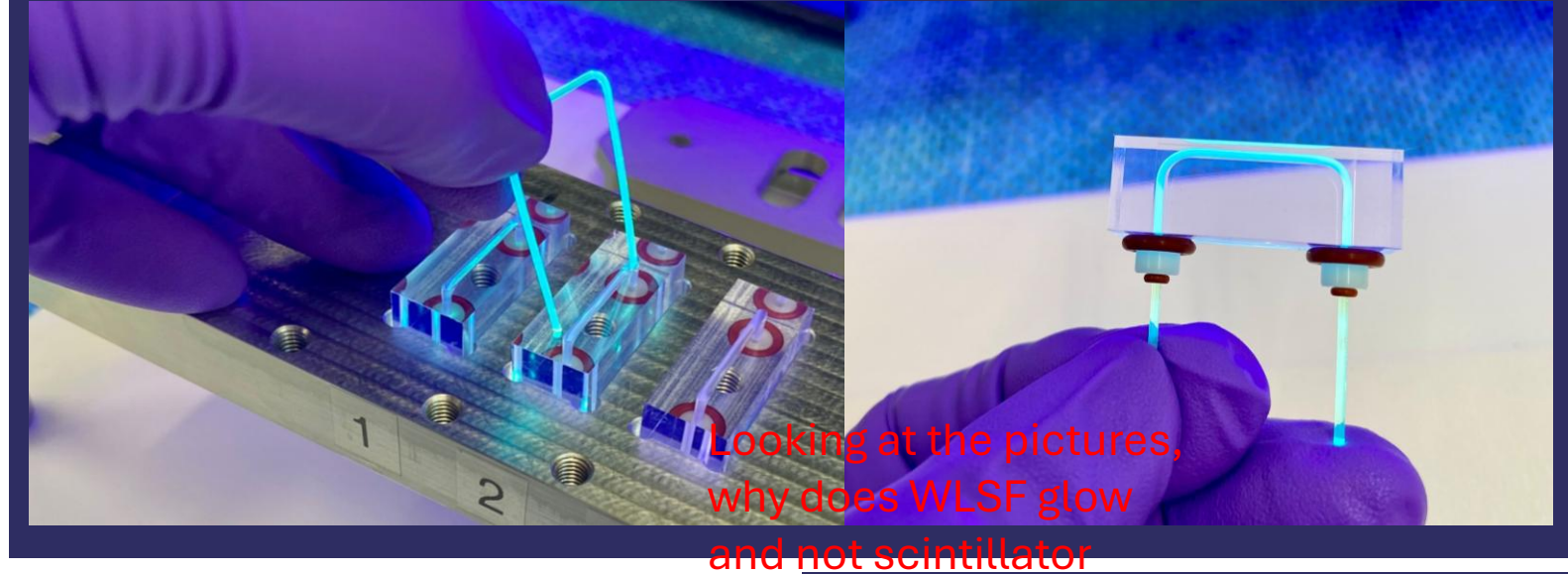
Studs for alignment

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- Pixelated detectors
- Geiger mode in each pixel (either on or off)
- Similar Gain to the PMT
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Tile Production Procedure

- Bend wavelength shifting fibre (WLSF)
- Insert WLSF into scintillator
- Insert studs
- Fix in place with degassed, transparent epoxy
- Cure the tile and set the resin
- Lap down protruding WLSF and stud
- Wrap in ESR
- Wrap with aluminum tape



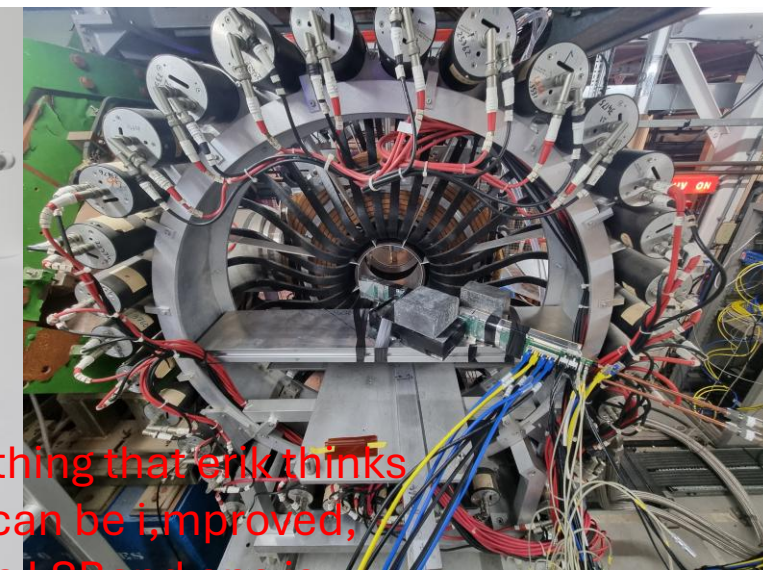
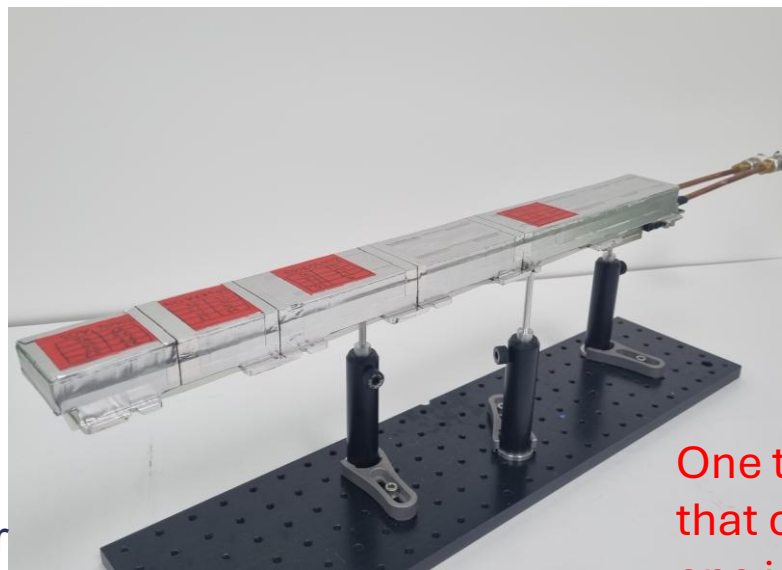
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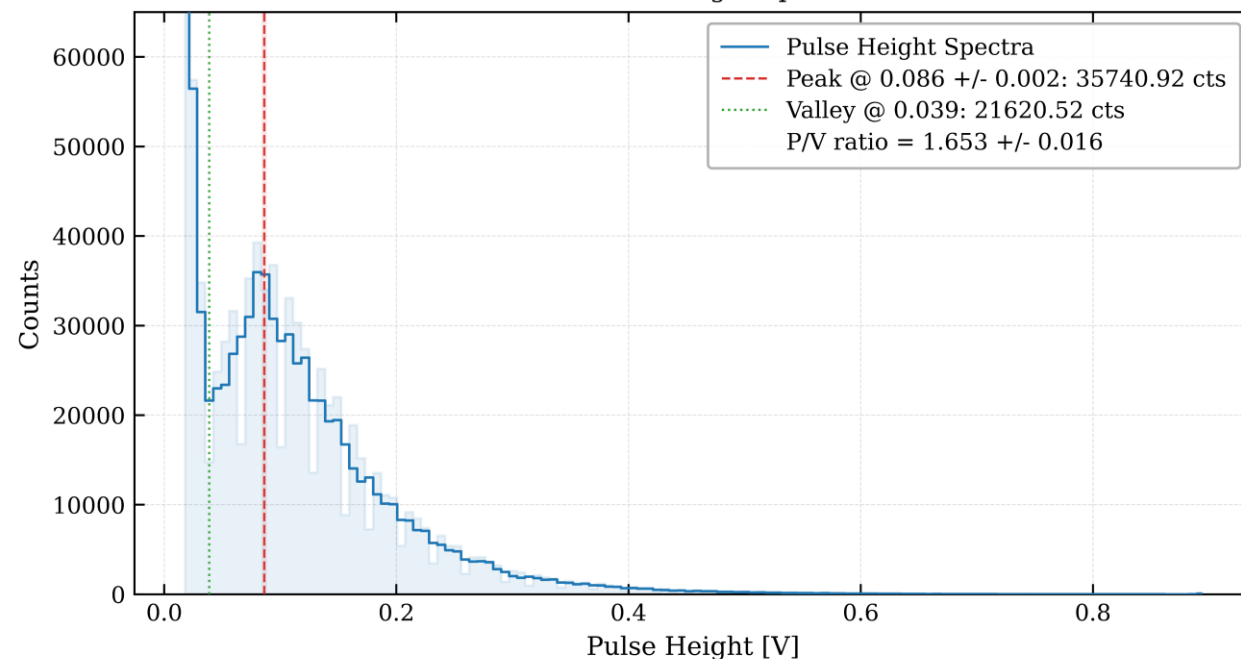
Determining Acceptance Threshold

- What makes a tile good enough for super MuSR?
- Requirement is, one stave must have an average P/V ratio exceeding that of super MuSR
- This was determined parasitically on the MuSR instrument
- MuSR P/V: 1.653 ± 0.016
- Super MuSR Average P/V = 2.715 ± 0.058
- All tiles exceeded performance except 210mm tiles 1.556 ± 0.133
- The tiles from this stave are suitable for final detector

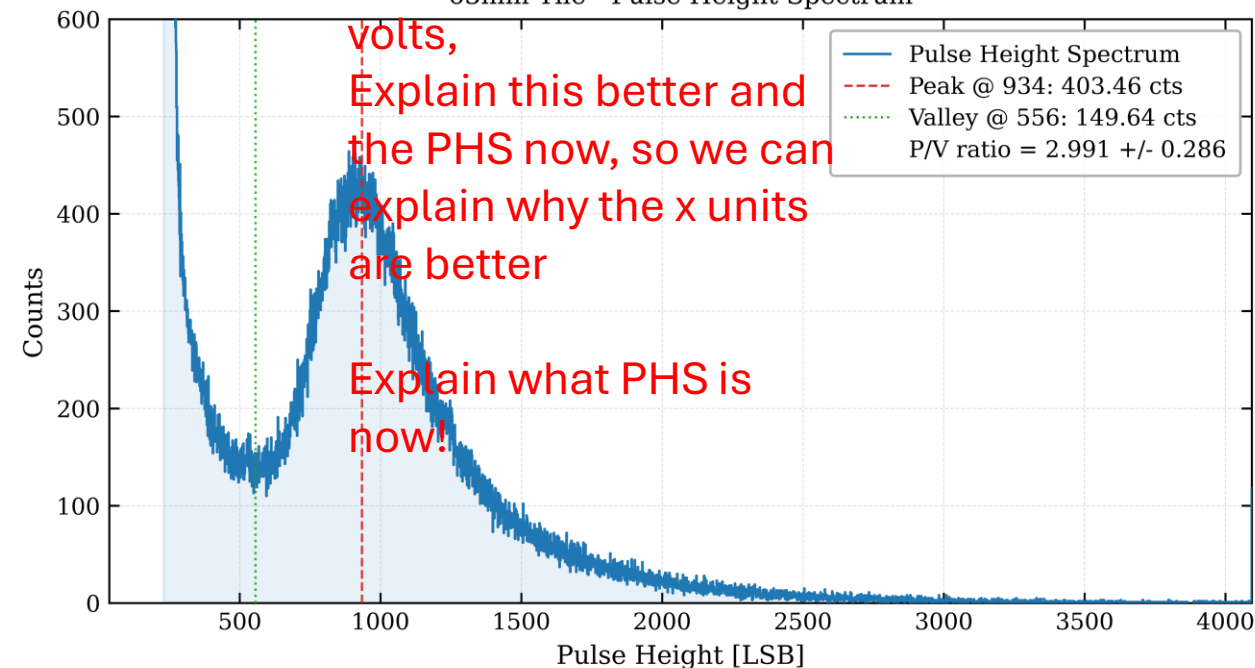


One thing that erik thinks that can be improved, one is LSBand one is volts,

MuSR Pulse Height Spectrum



63mm Tile - Pulse Height Spectrum

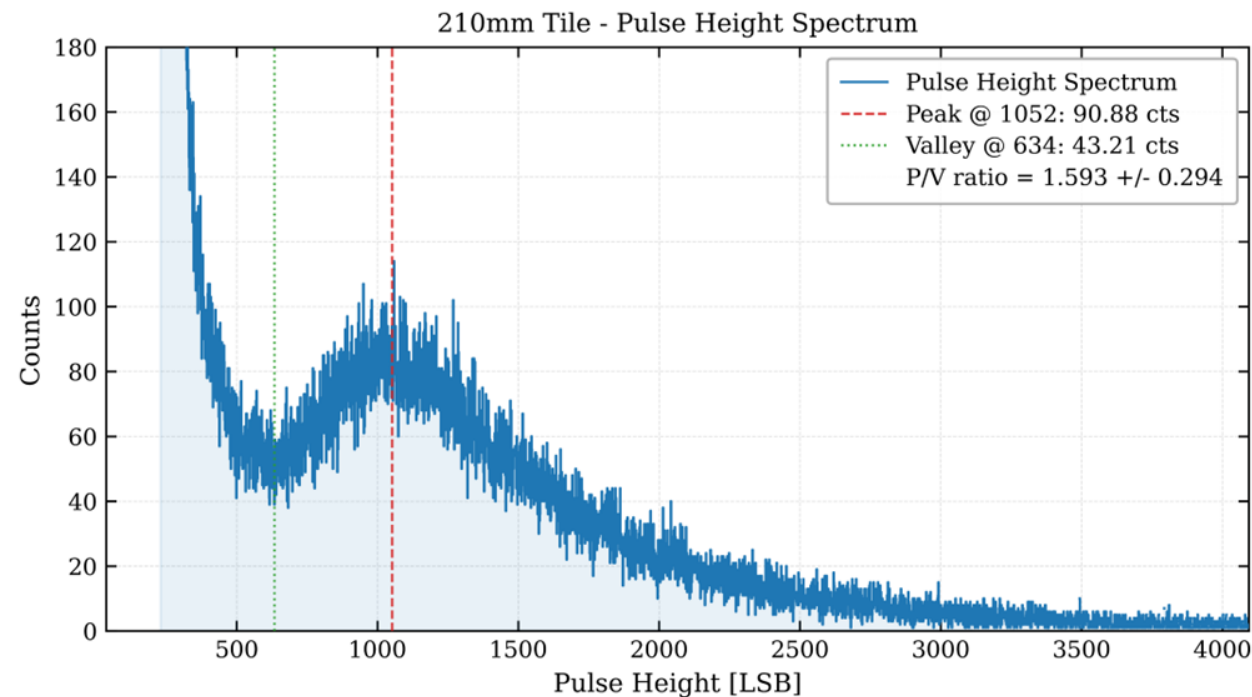
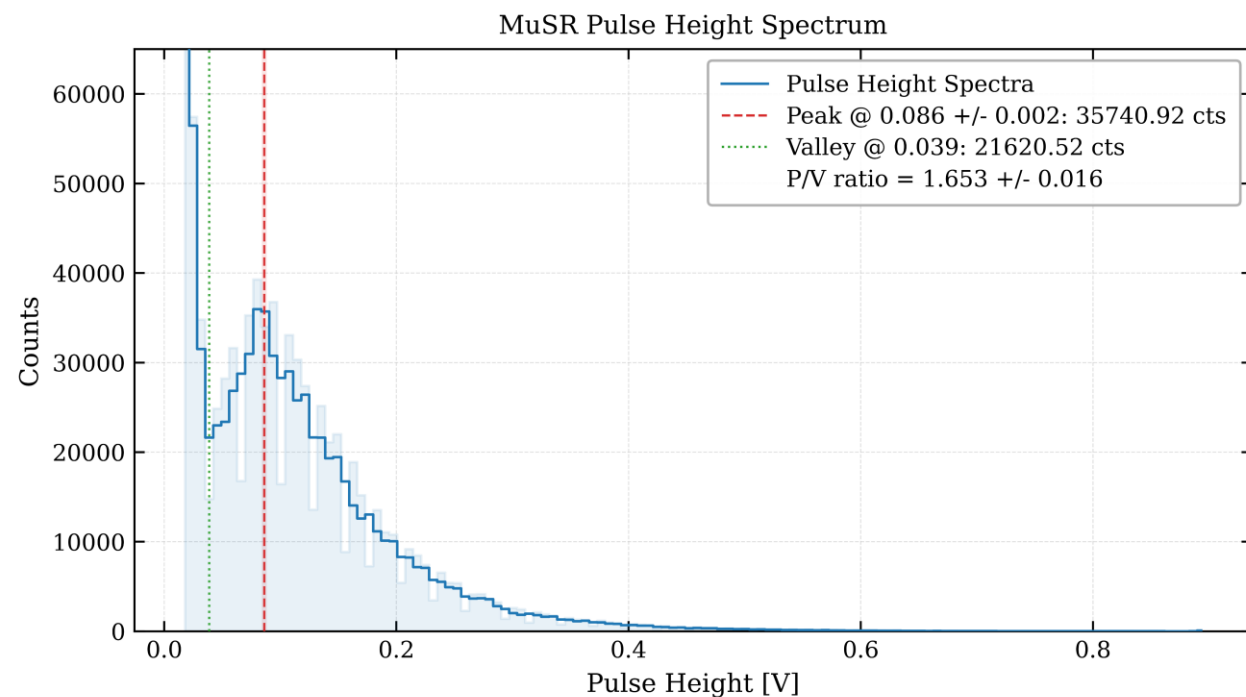
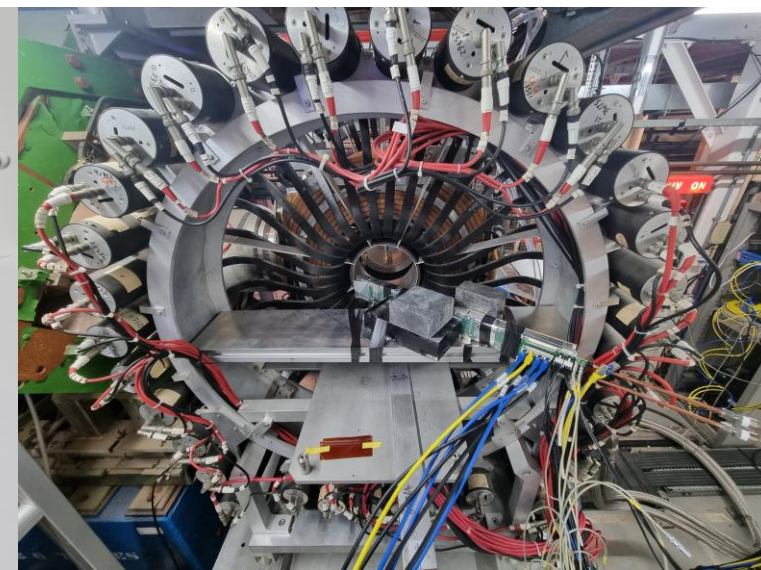
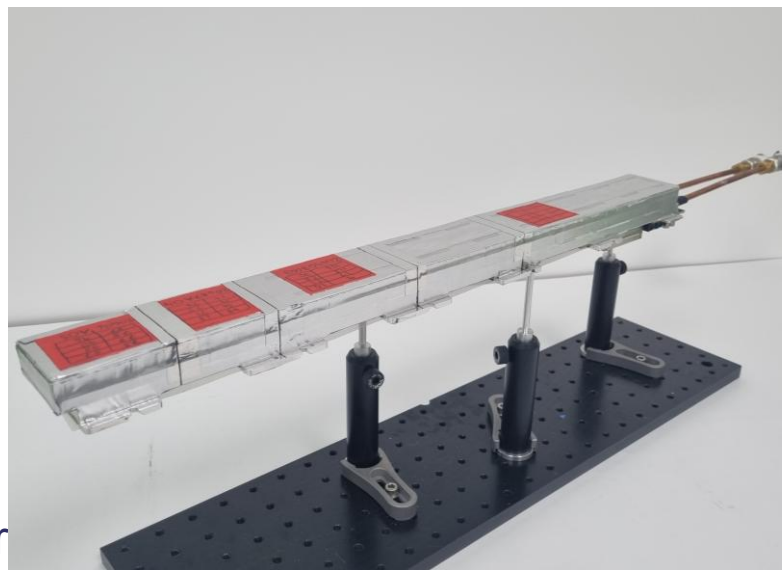


Explain this better and the PHS now, so we can explain why the x units are better

Explain what PHS is now!

Determining Acceptance Threshold

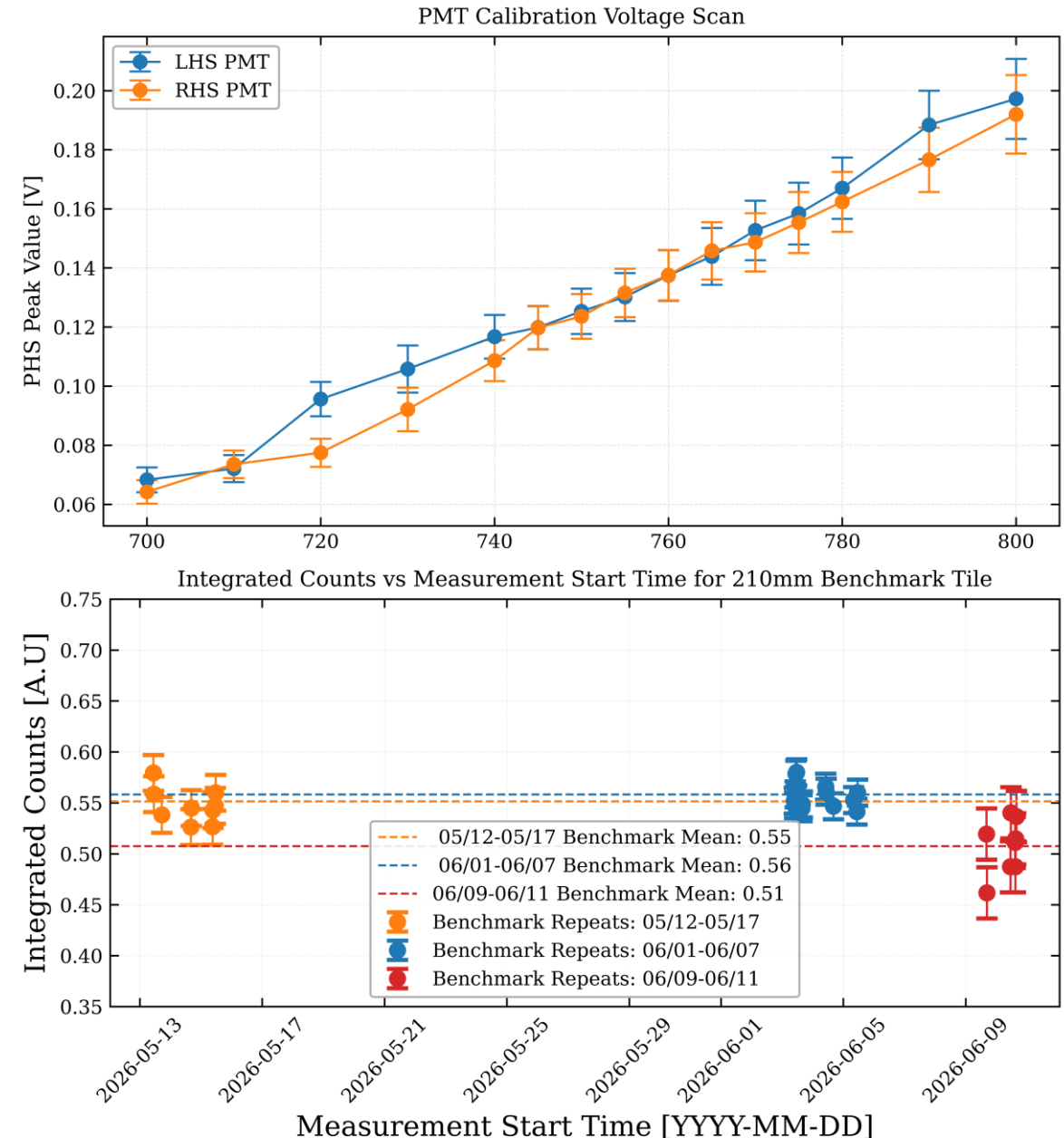
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Tile Measurement Calibration & Error

- Setup calibrated by doing a gain scan and matching both PMTs
- Error dominated by optical coupling and source alignment
- On a longer timeframe, error also arises due to PMT gain drift, ambient temperature changes etc.
- Both can be solved by taking many measurements of the benchmark tile for each length.
- Gives estimate of error per measurement with std deviation of repeats (error from experimental setup) within a batch of measurements
- Allows for each batch of measurements to be normalized relative to the benchmark tile, allows for comparison between batches

Change slides 13 and 12

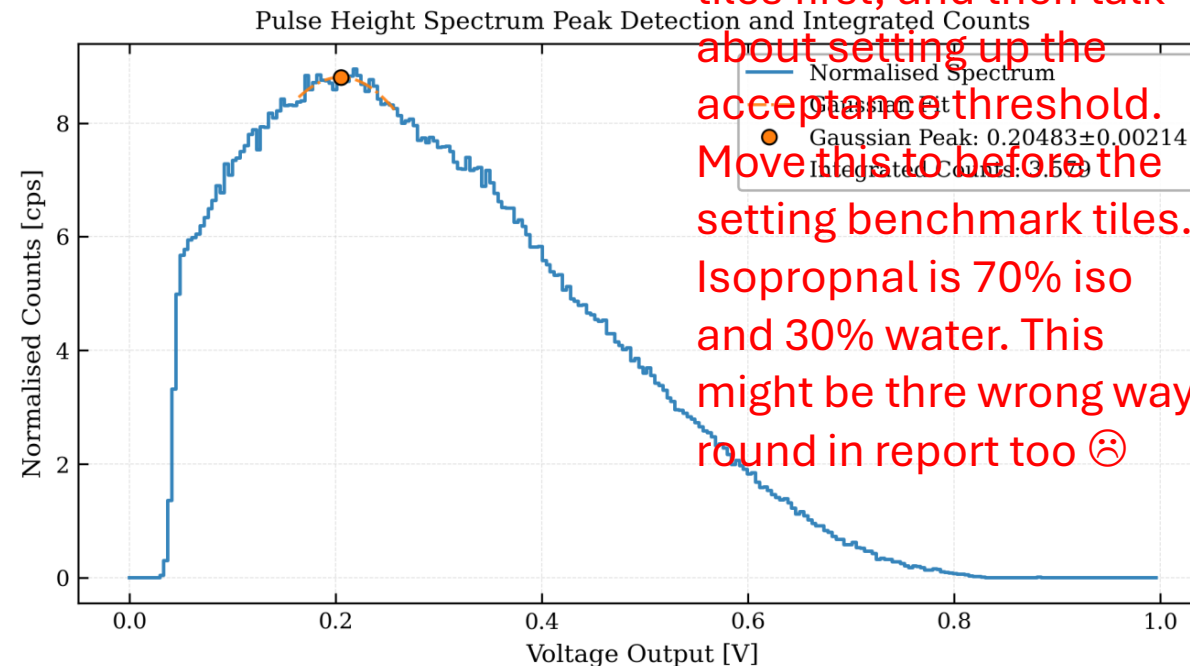


Tile Measurement Procedure

- We need to compare the verified tiles, to ones not tested on the beamline.
- To determine light output of the tiles PMTs were used.
- Clean tile and PMTs with 70% denoised water 30% ISO
- Apply optical vacuum grease to bottom of the studs
- Align Sr90 Source with the centre of the tile
- Collect total PHS
- Determine light output FoM using either Integrated counts or PHS peak



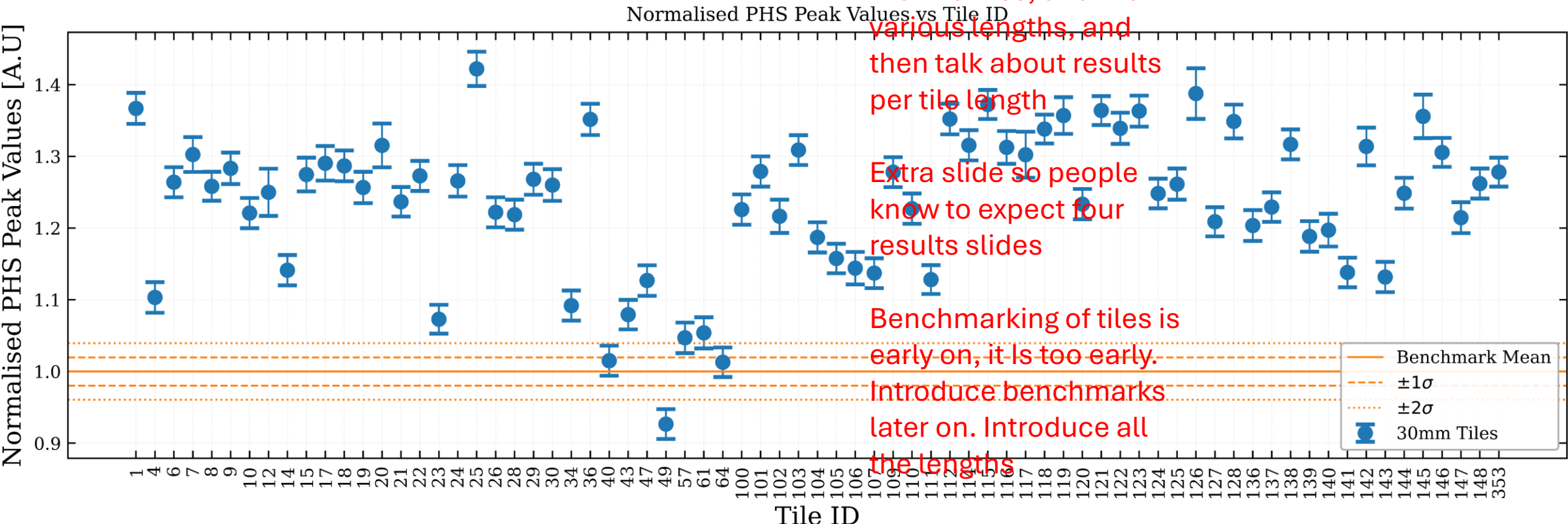
Move this slide earlier, explain how we measure tiles first, and then talk about setting up the acceptance threshold. Move this to before the setting benchmark tiles. Isopropyl is 70% iso and 30% water. This might be the wrong way round in report too 😞



30mm Tiles

- Benchmark tile was determined in previous work, so all tiles must exceed light output. 73 tiles tested overall
- 23.7% better overall, 1 tile falls below the acceptance threshold (1.4% of measured tiles)
- Tiles within 2 standard deviations were investigated
- Initially used IN2 resin, cloudy finish led to it being replaced by RAL71 resin
- Relapping & resin type investigated as cause

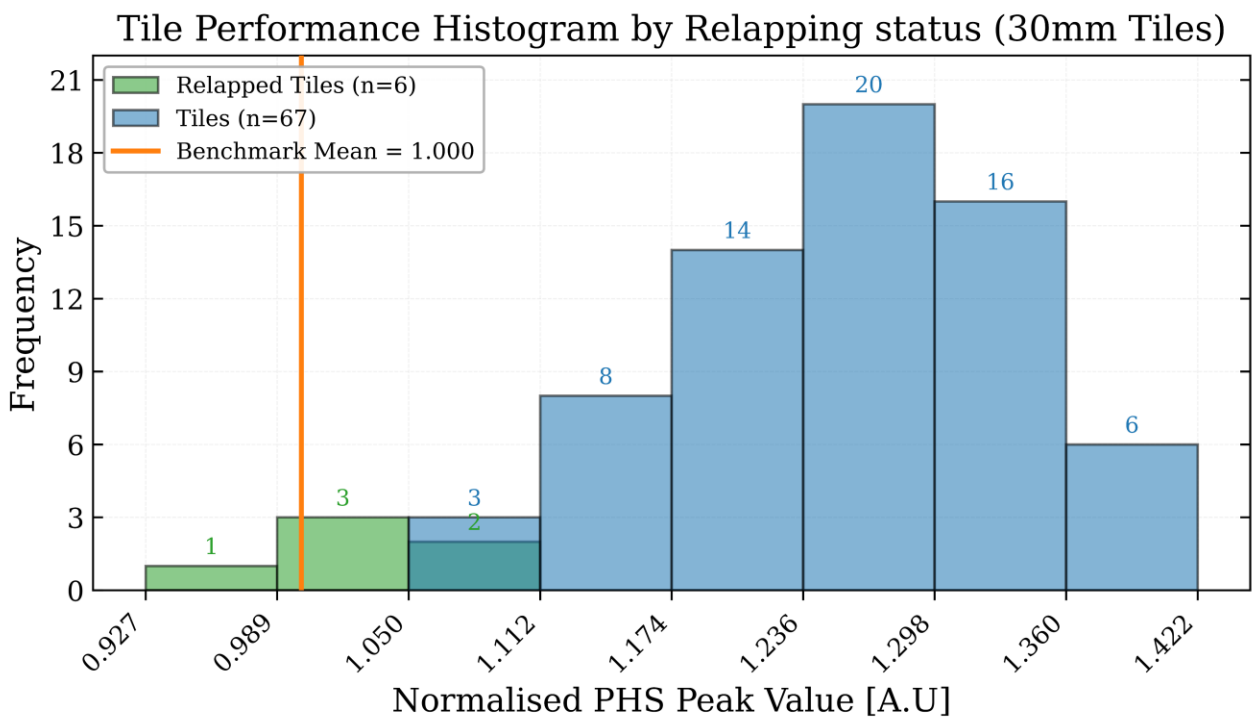
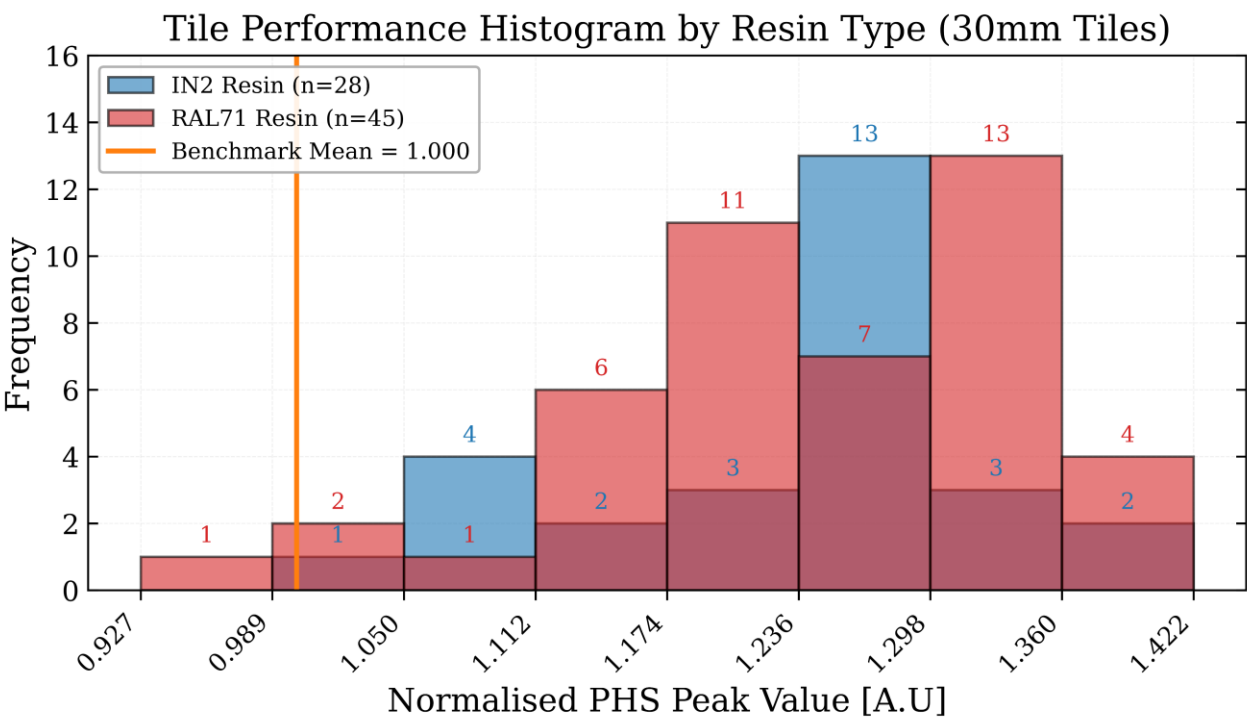
ID	Resin	Production Notes
40	IN2	Divot both ends, dull crackly finish, bubble one corner, re-lapped
49	RAL71	Clear resin, bubble on one fibre leg, Some small scratches on tile, re-lapped
57	RAL71	Heavily washed tile, clear resin, bubble in 1 stud, mark on tile, re-lapped
61	RAL71	Clear resin, bubble/air in one leg, re-lapped
64	RAL71	Clear resin, possible contamination in both legs, bubble in one stud, re-lapped
120*	RAL71	Tiny bubble in 1 stud and red spec in the other



30mm Tiles

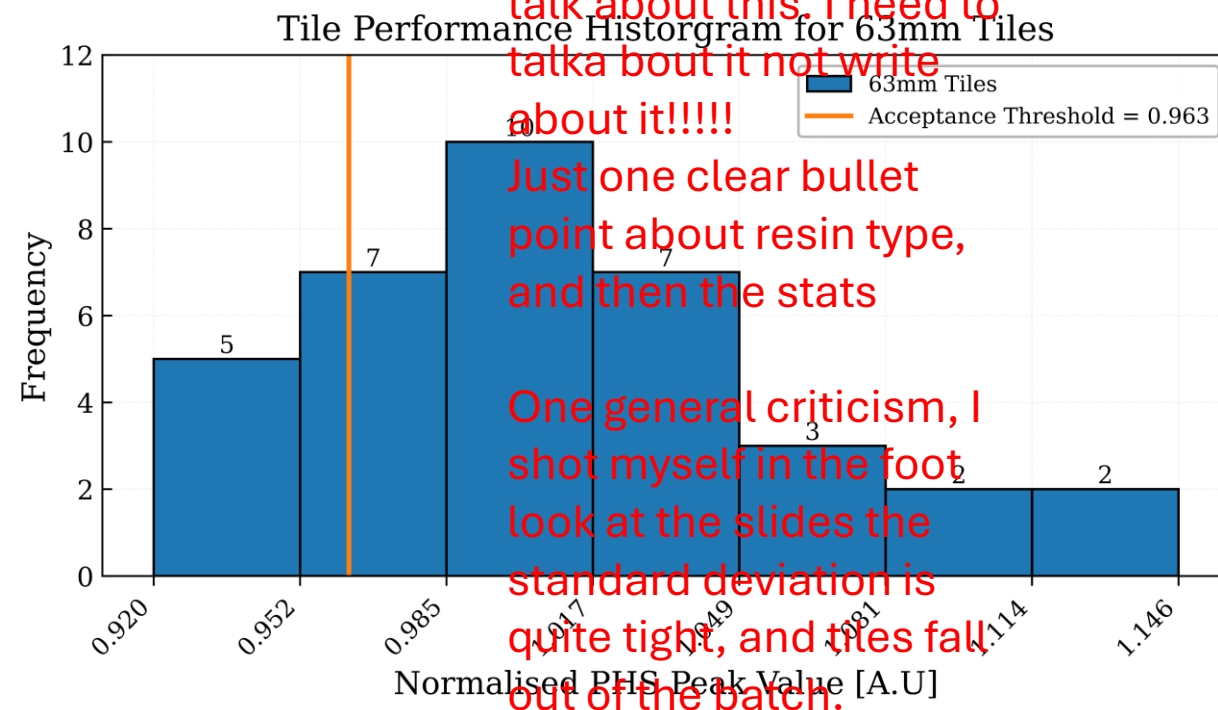
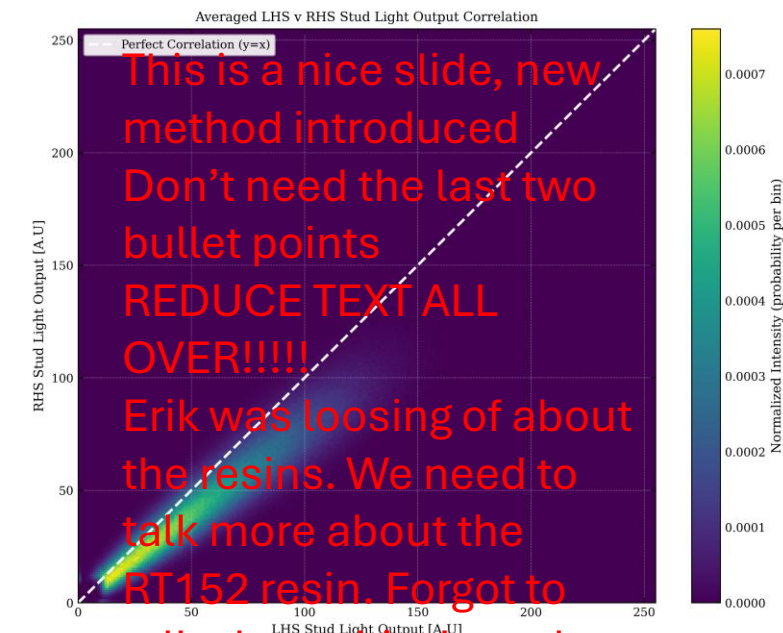
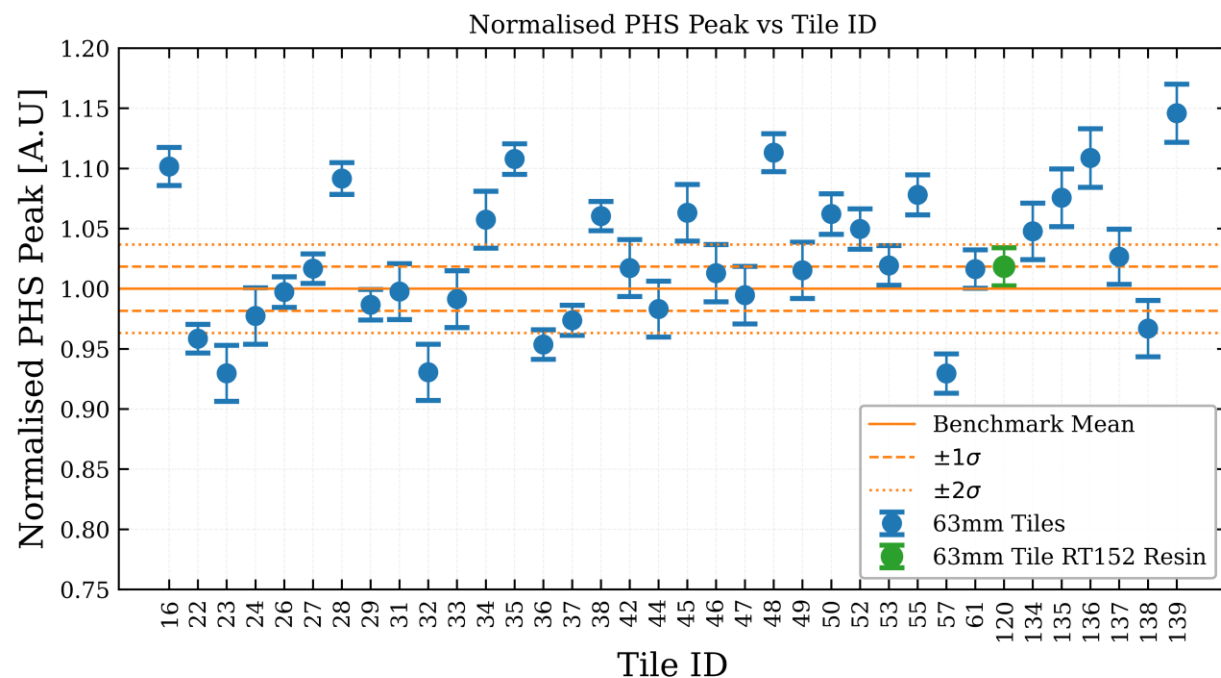
- Separated Tile by resin type, no statistically significant difference in tile performance when split by resin type giving a not significant result (T-test $p=0.756$)
- re-lapped tiles are performing worse (T-test $p=1.273 \times 10^{-9}$), statistically significant.
- Only 6 tiles were re-lapped, but no evidence single outlier caused this result.
- Negative skew due to tiles being produced near max light output

ID	Resin	Production Notes
40	IN2	Divot both ends, dull crackly finish, bubble one corner, re-lapped
49	RAL71	Clear resin, bubble on one fibre leg, Some small scratches on tile, re-lapped
57	RAL71	Ultrasonic washed tile, clear resin, bubble in 1 stud, mark on tile, re-lapped
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64	RAL71	Clear resin, possible contamination in both legs, bubble in one stud, re-lapped
120*	RAL71	Tiny bubble in 1 stud and red spec in the other



63mm Tiles

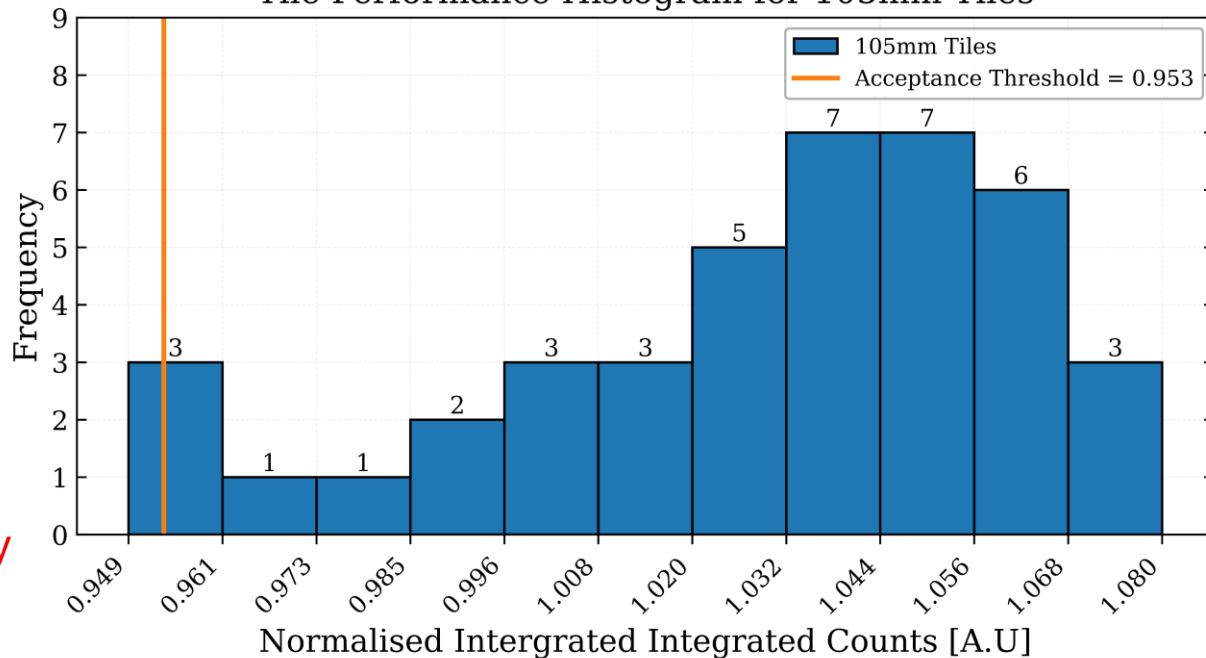
- 36 tiles were tested, 13.9% fell below the acceptance threshold
- damage to studs was noted in the technicians' notes
- New diagnostic tool used for the longer tiles, we can do a correlation plot between the LHS and RHS pulses, which can indicate asymmetry in the tile, due to stud damage, poor bonding between WLSF and scintillator on one side of the tile.
- 63mm tiles were used to test a new resin type RT152, worry about longer tiles deforming during curing process, trials with this resin for 63mm tiles verified use for the longer tiles
- We don't see a negative gaussian because already refined production process (0.055 vs 0.100 standard deviation for 30mm and 63mm tiles respectively)
- Tiles fell below randomly, no statistically significant result for any given defect



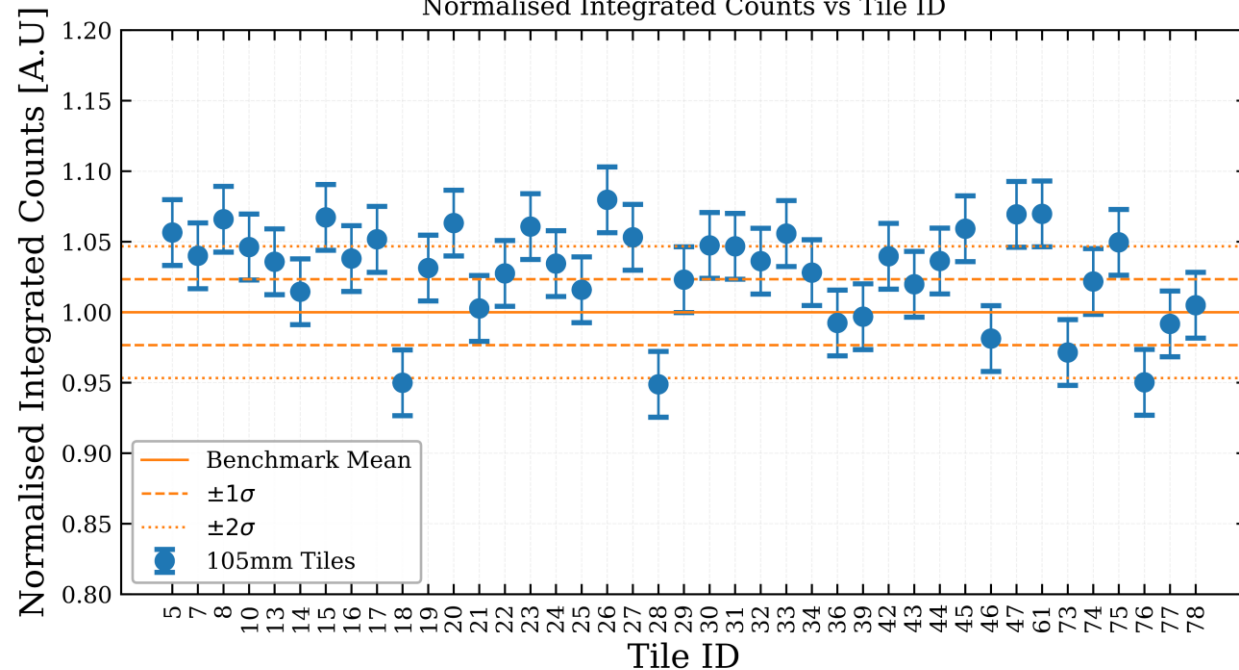
105mm Tiles

- 41 Tiles were tested, overall, 2.9% better performance than the benchmark tile repeats
- 3/41 tiles were flagged for investigation as they fell below the benchmark mean
- No single fault was identified, but the negatively skewed gaussian is recovered
- This randomly across the whole tile production process, this is a result of the inherent difficulty of manufacturing the longer tiles
- Longer tiles are more susceptible to errors in manufacturing as they are less bright
- All fall within 5% of the benchmark mean, further tests to be conducted to determine if they are viable for the final detector

Tile Performance Histogram for 105mm Tiles

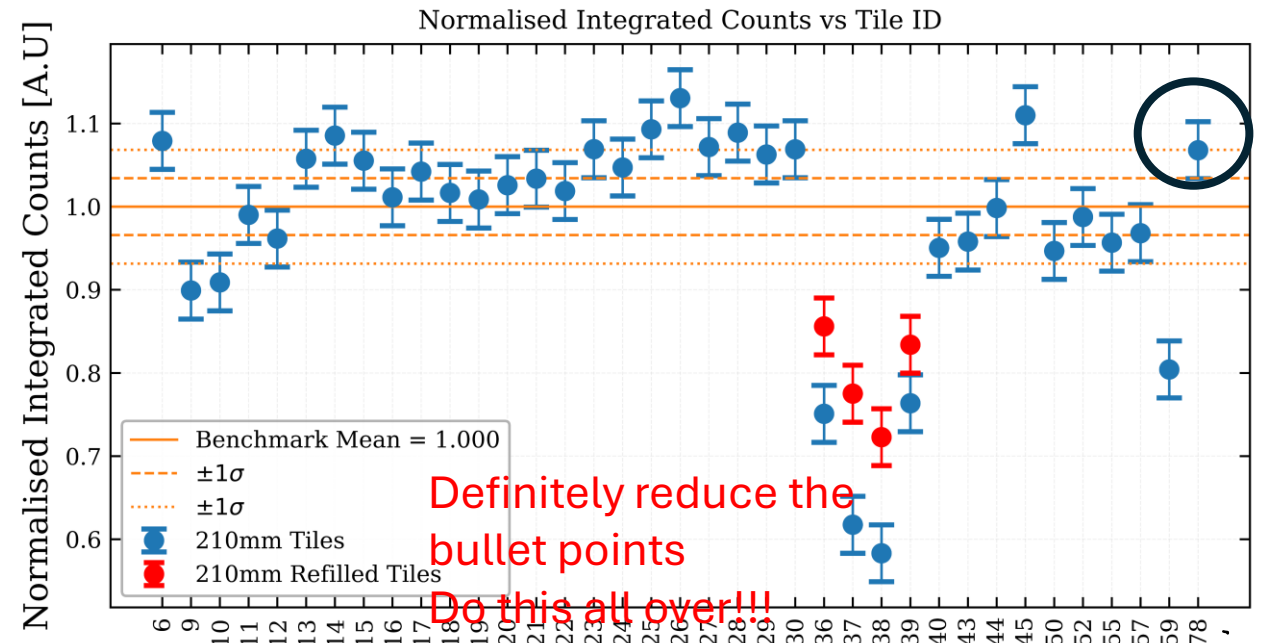
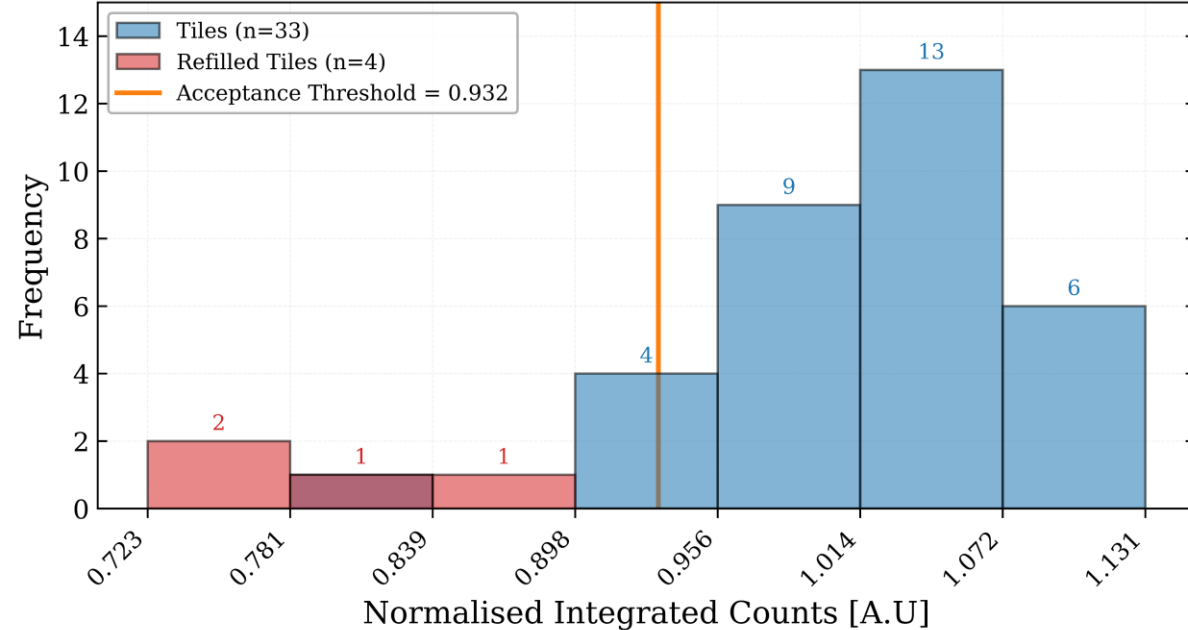


Normalised Integrated Counts vs Tile ID



210mm Tiles

Integrated Count distribution by Refill status (210mm Tiles)



Definitely reduce the bullet points

Do this all over!!!

Took erik a while for Id

79, did I make a typo

potentially highlight it.

Erik could not find tile

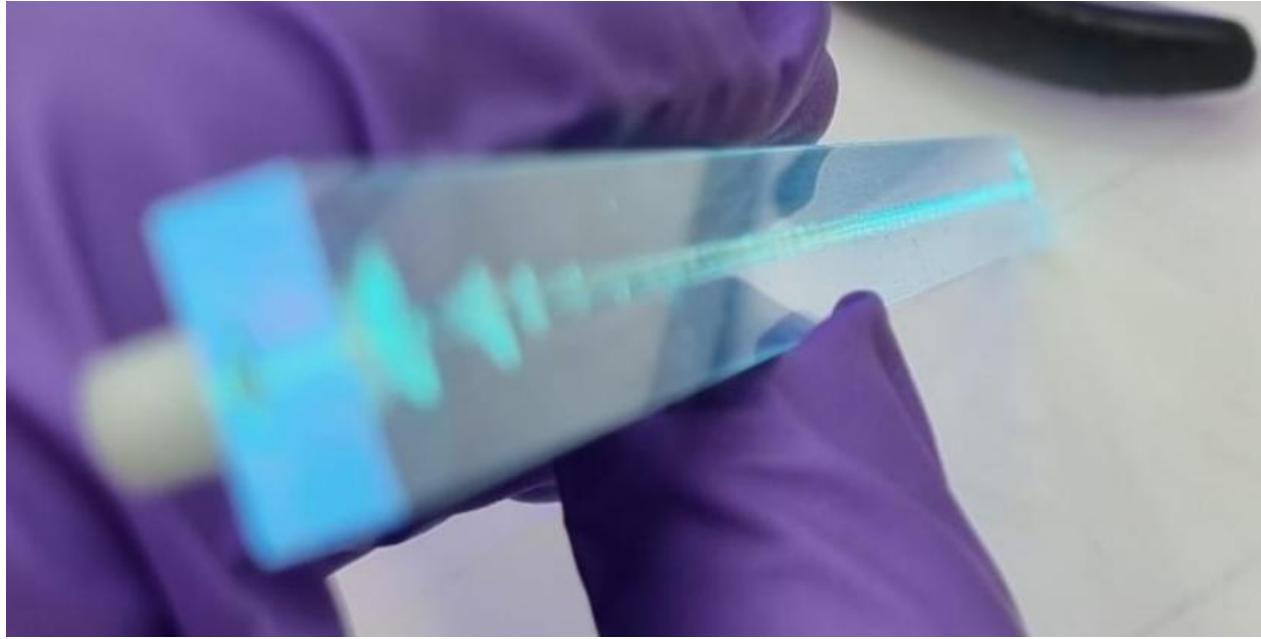
78.

TOO MANY BULLET POINTS

- 37 Tiles tested were tested, 18.9% of the tiles fell below the acceptance threshold
- A group of tiles from the same batch were driving this failure rate (32.1% decrease from the benchmark mean), so they were investigated.
- No clear indication in the technicians notes on a flaw, so they were unwrapped, it was found that they did not have suitable resin applied during the tile testing process. This is theorised to be causing a poor optical connection between the WLSF and scintillator
- Applying more resin increased brought the tiles to 20.3% below the benchmark mean
- Did not fully recover performance, unable to fill in some gaps, and contamination between resin surfaces
- Re introduced adding resin such that it pools along the top of tile ensuring strong optical connection, ID 78 show was done with this method and shows tile performance is back to above the acceptance threshold.
- Excluding the refilled tiles, 8.8% fall below the acceptance threshold.

Resin Pooling Issues

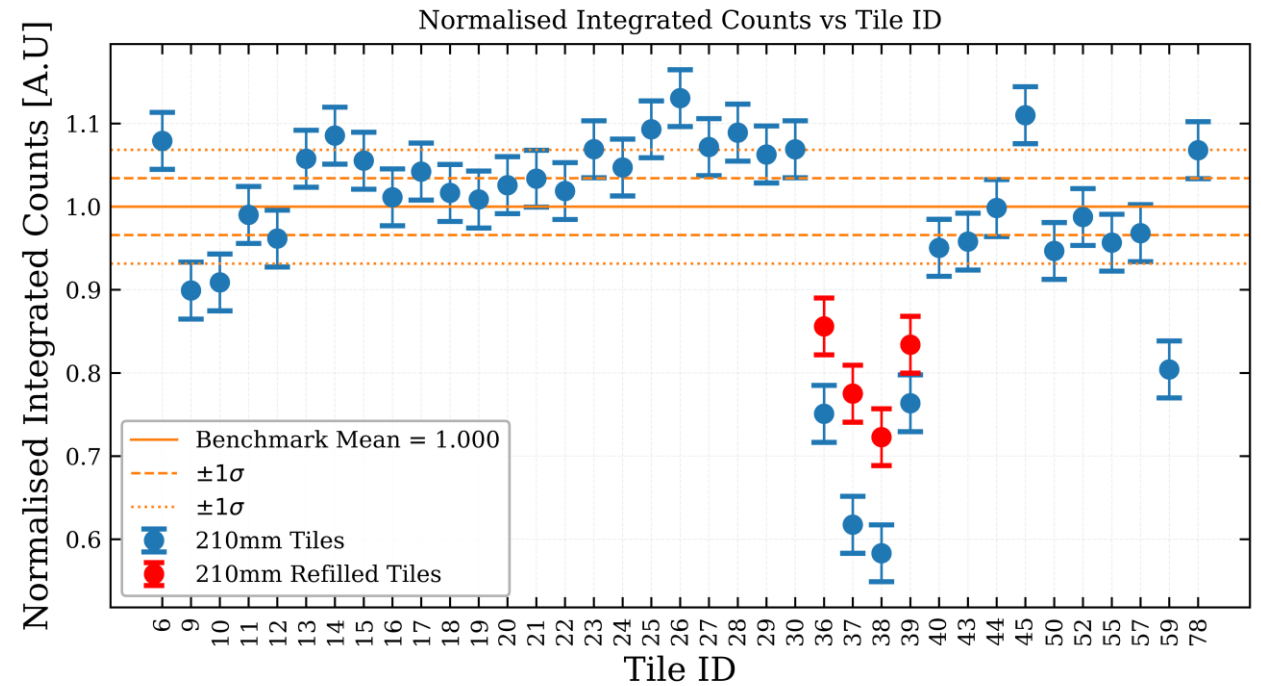
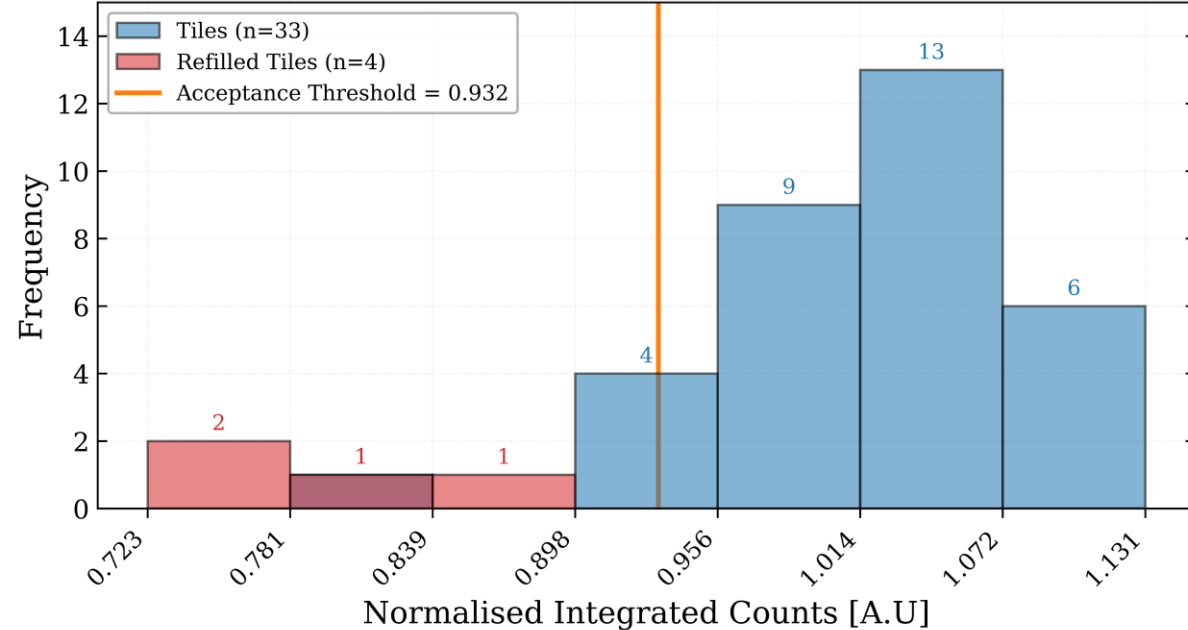
Extremely smooth, no resin pooling along the top of the tile



Lots of visible pooling across the top of the tiles

210mm Tiles

Integrated Count distribution by Refill status (210mm Tiles)



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Conclusions & Future Work

Conclusions

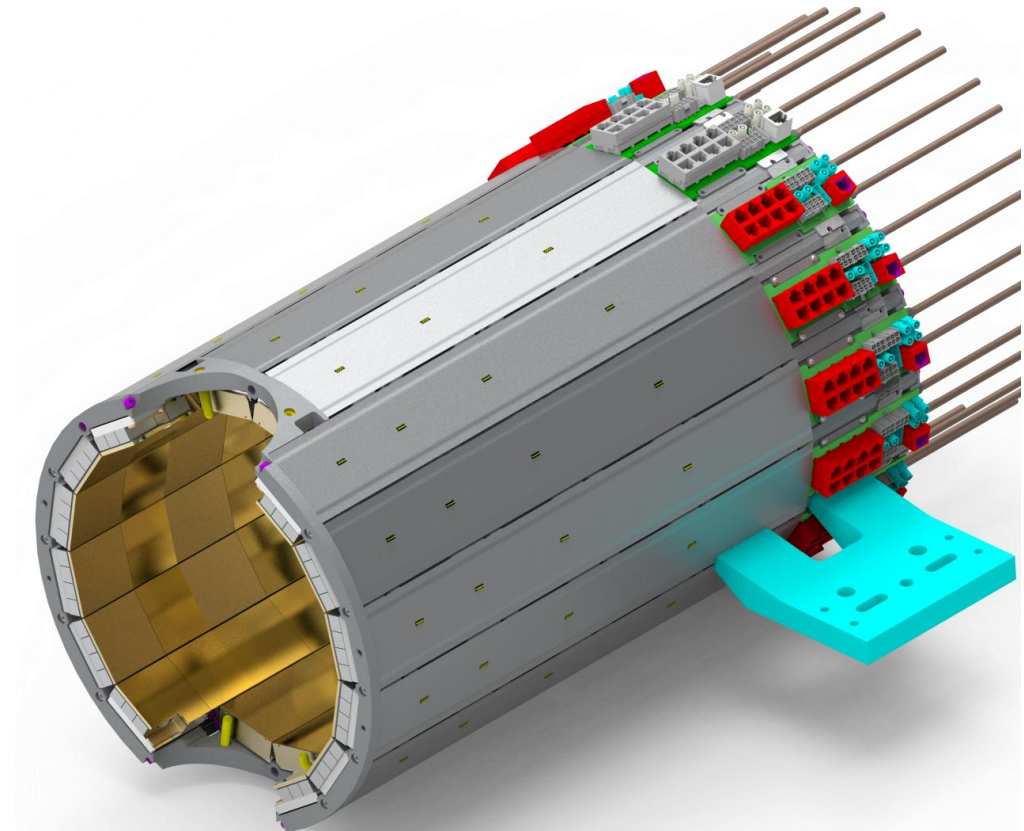
- Main causes of damage throughout all tiles appear to be re-lapping and poor optical coupling
- 8.3% of all tiles flagged for investigation

Future work

- Testing 43mm tiles
- Develop absolute test of brightness, as current tests are comparative
- Flood field testing and beamline tests, whole modules to be tested simultaneously
- Continue to monitor tiles to ensure they are of high quality for the Super MuSR final build

Notes from erik

- Start with positive thing for the first point of the tiles
- Maybe mention how many tiles we proved are good
- Frame more positively include total number of tiles tested in this series of measurements
- Mention that it is a conservative estimate, most of these will probably pass
- Quite an abrupt conclusion
- Lots of info but “cant see the root for the trees”





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Thank you

GENERAL CRITISIM:
MORE TALKING LESS
BULLET POINTS

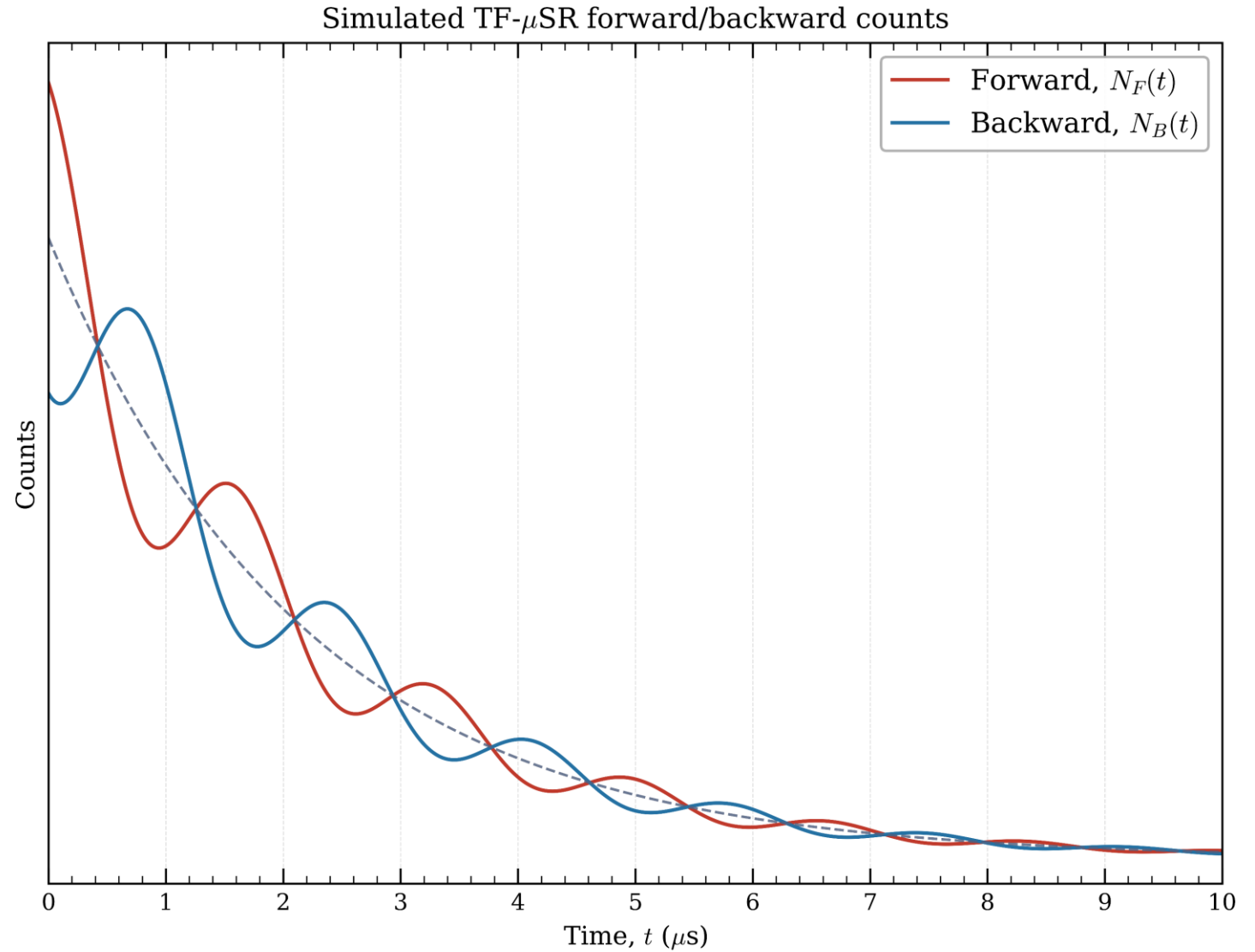


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Questions?

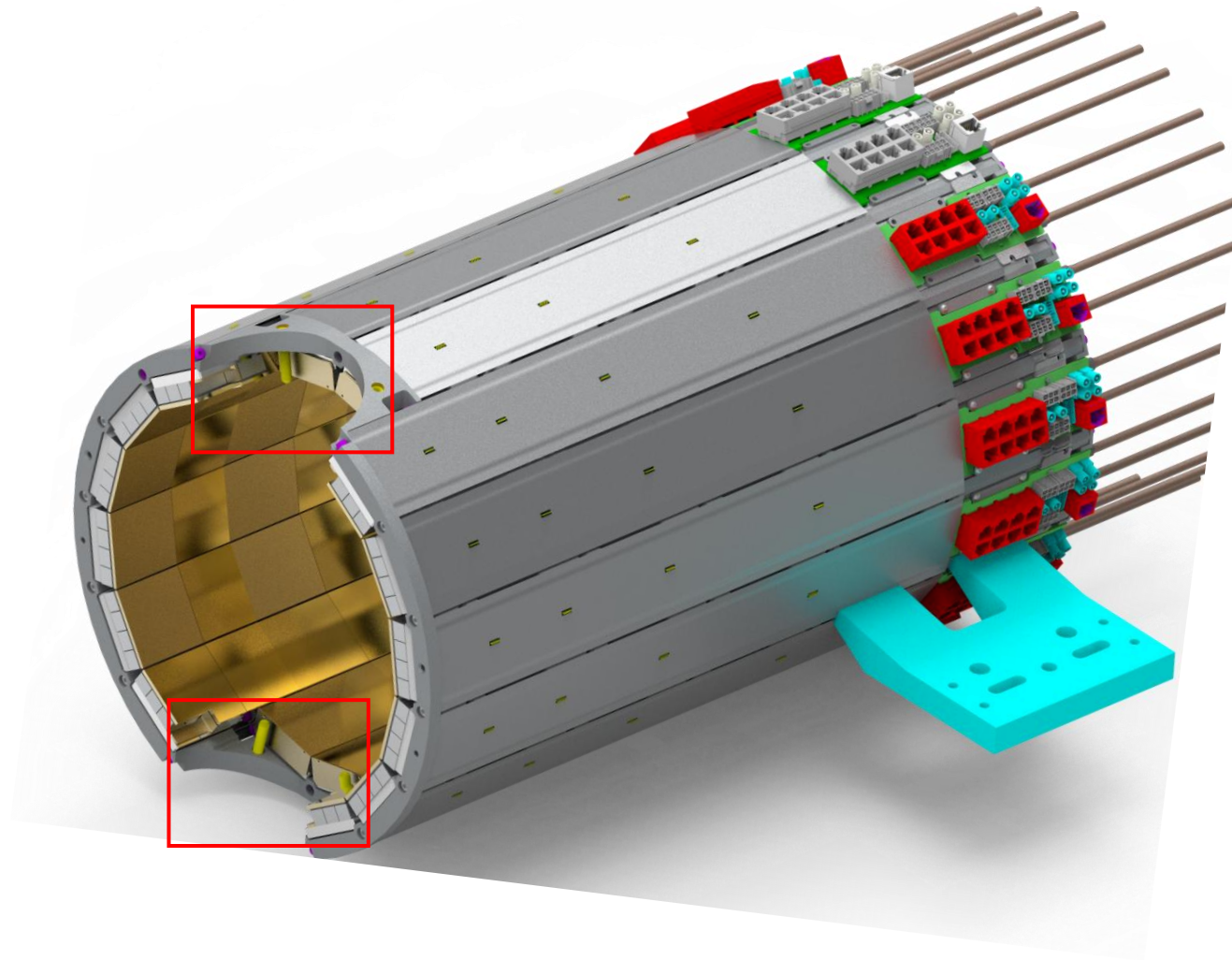
Asymmetry equation

$$A(t) = \frac{N_F(t) - \alpha N_B(t)}{N_F(t) + \alpha N_B(t)} = a_0 P_z(t)$$



Why not 1024 detectors?

- To make room for the sample and cryostat to be lowered into the detector, 64 detectors have to be removed from each detector.
- This leaves 960 detectors remaining



Why do we switch to integrated counts?

- Getting PHS peak finder to fit to the 105mm and 210mm tiles was very difficult.
- Integrated counts gives an accurate “overall light output”

